DATA 624 Homework 5

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```r
library(readxl)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(ggplot2)
library(fpp2)
```

```
## Warning: package 'fpp2' was built under R version 4.5.2

## -- Attaching packages ---------------------------------------------- fpp2 2.5.1 --

## v fma       2.5     v expsmooth 2.3

##
```

Introduction

In this report, I will be answering questions 1, 5, 6, 7, and 8 in the 8th chapter from the Hyndman book.

Question 1

Figure 8.31 shows the ACFs for 36 random numbers, 360 random numbers and 1,000 random numbers.

(a) Explain the differences among these figures. Do they all indicate that the data are white noise?

The three ACF plots look different mainly because of the different sample sizes. The series with 36 observations looks much noisier, with larger mins and maxs, since small samples produce more random variation in the sample autocorrelations. As the sample size increases to 360 and then 1,000 observations, the ACF values stay closer to zero and the plots look more stable. Even though the plots look different, all three are consistent with white noise since none of them show a clear pattern (increasing, decreasing, or seasonal structure). In fact, the autocorrelations fluctuate randomly around zero, which is what is expected from a white noise process.

(b) Why are the critical values at different distances from the mean of zero? Why are the autocorrelations different in each figure when they each refer to white noise?

The critical values are at different distances from zero because they depend on the sample size. The approximate critical values for white noise are + or - 1.96/√T, where T is the number of observations. When the sample size is small, there is more sampling variability. As the sample size increases, the bounds become narrower because the estimates become more precise. Each plot comes from a different random sample, which makes the white noise representations to be different. In theory, white noise has zero autocorrelation at all lags, but in practice the estimated autocorrelations will vary slightly due to random sampling. Larger samples produce estimates that are closer to zero.

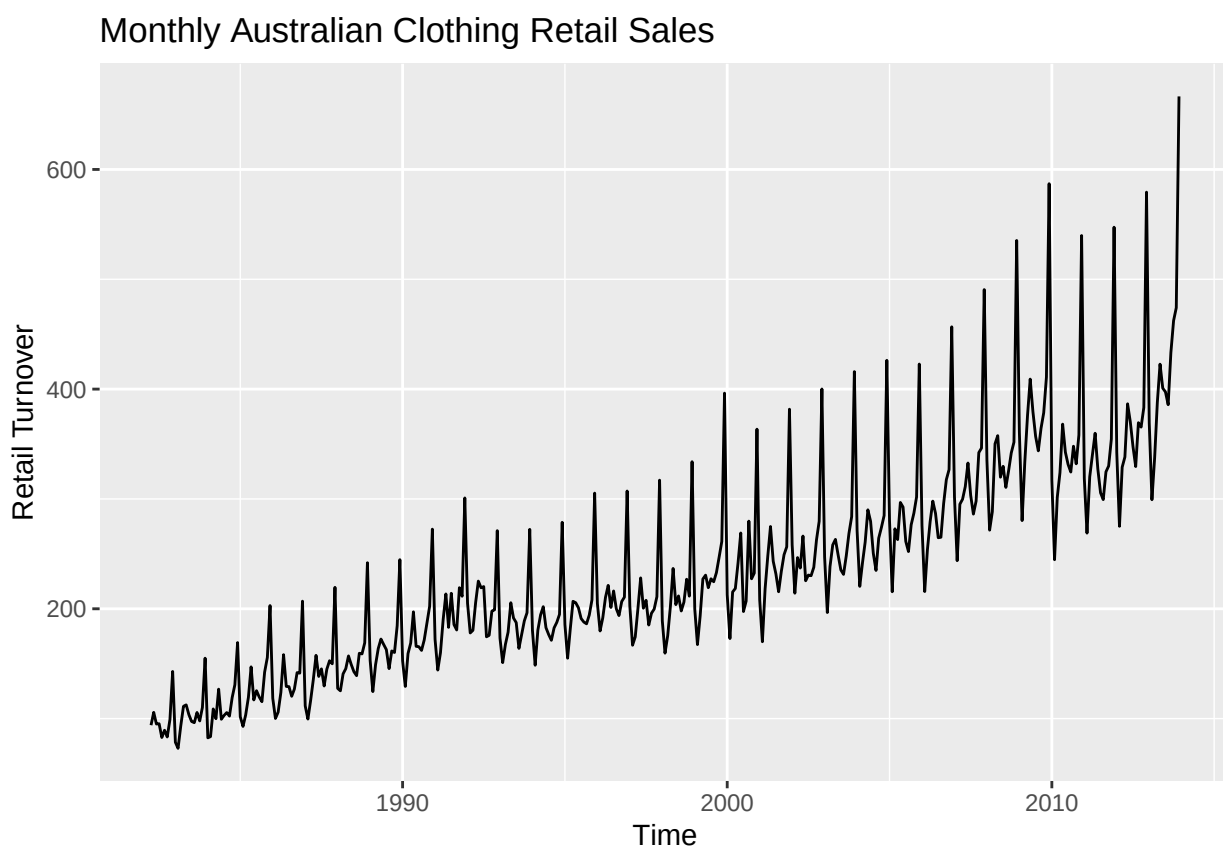
Question 5

For your retail data (from Exercise 3 in Section 2.10), find the appropriate order of differencing (after transformation if necessary) to obtain stationary data.

```
retaildata_raw <- "https://raw.githubusercontent.com/JDO-MSDS/Data-624/refs/heads/main/Homework%205/reta
retaildata <- read.csv(retaildata_raw, skip = 1)
```

```
myts <- ts(retaildata[,"A3349399C"],
           frequency = 12,
           start = c(1982,4))
```

```
autoplot(myts) +
  labs(title = "Monthly Australian Clothing Retail Sales",
       y = "Retail Turnover")
```



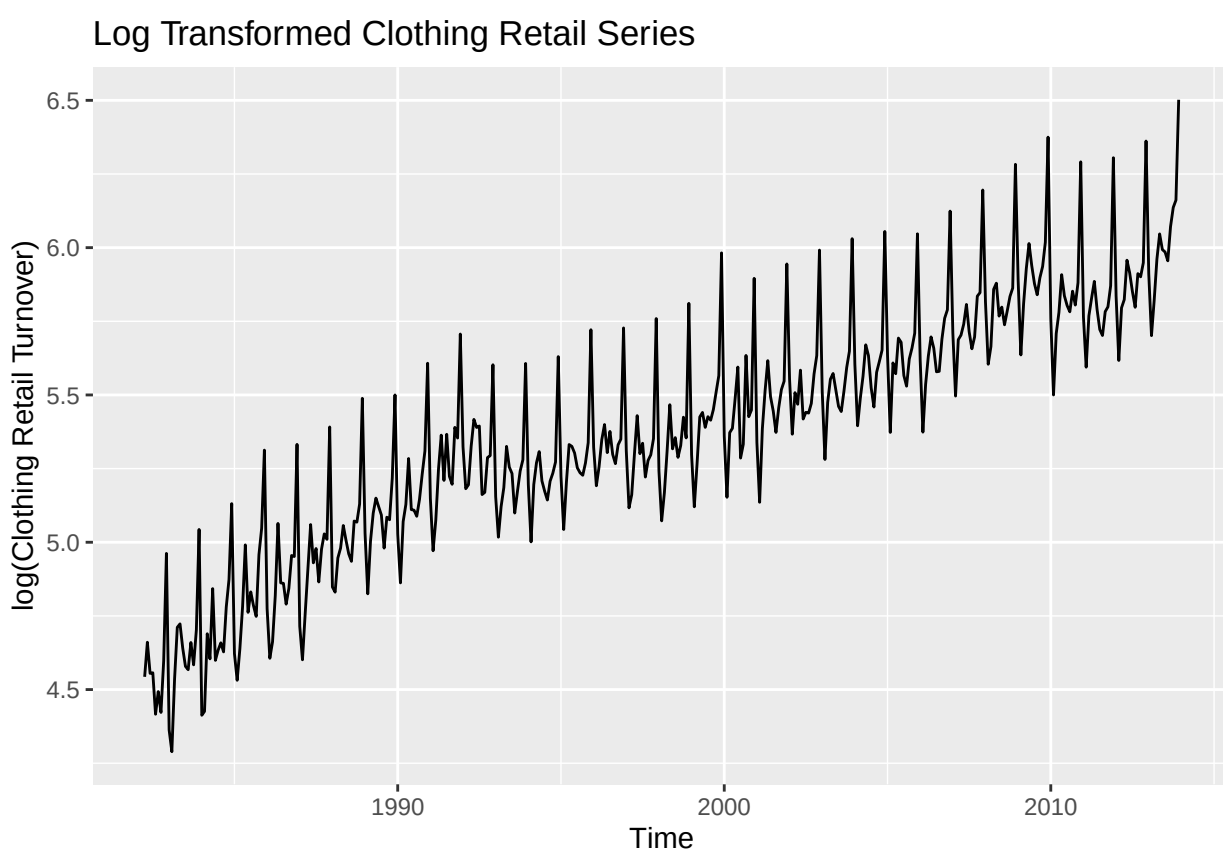
Trasformation needed?

```r
lambda <- BoxCox.lambda(myts)
lambda
```

```
## [1] 0.02074707
```

Since lambda is 0.02074707, which is very close to zero, I will apply a log transformation.

```r
myts_trans <- log(myts)
autoplot(myts_trans) +
  labs(title = "Log Transformed Clothing Retail Series",
       y = "log(Clothing Retail Turnover)")
```



The original retail series shows an upward trend and strong seasonal patterns. The seasonal extremes become larger as the level of the series increases, indicating that the variance grows over time. Because of this, I considered a Box-Cox transformation. The estimated parameter was 0.0207, which is very close to zero. This suggests that a log transformation is appropriate.

After applying the log transformation, the seasonal fluctuations seem a little more stable and the variance is more consistent over time. However, the transformed series still shows the same general trend and seasonality, but the scale is more suitable for further time series analysis.

```r
ndiffs(myts_trans)
```

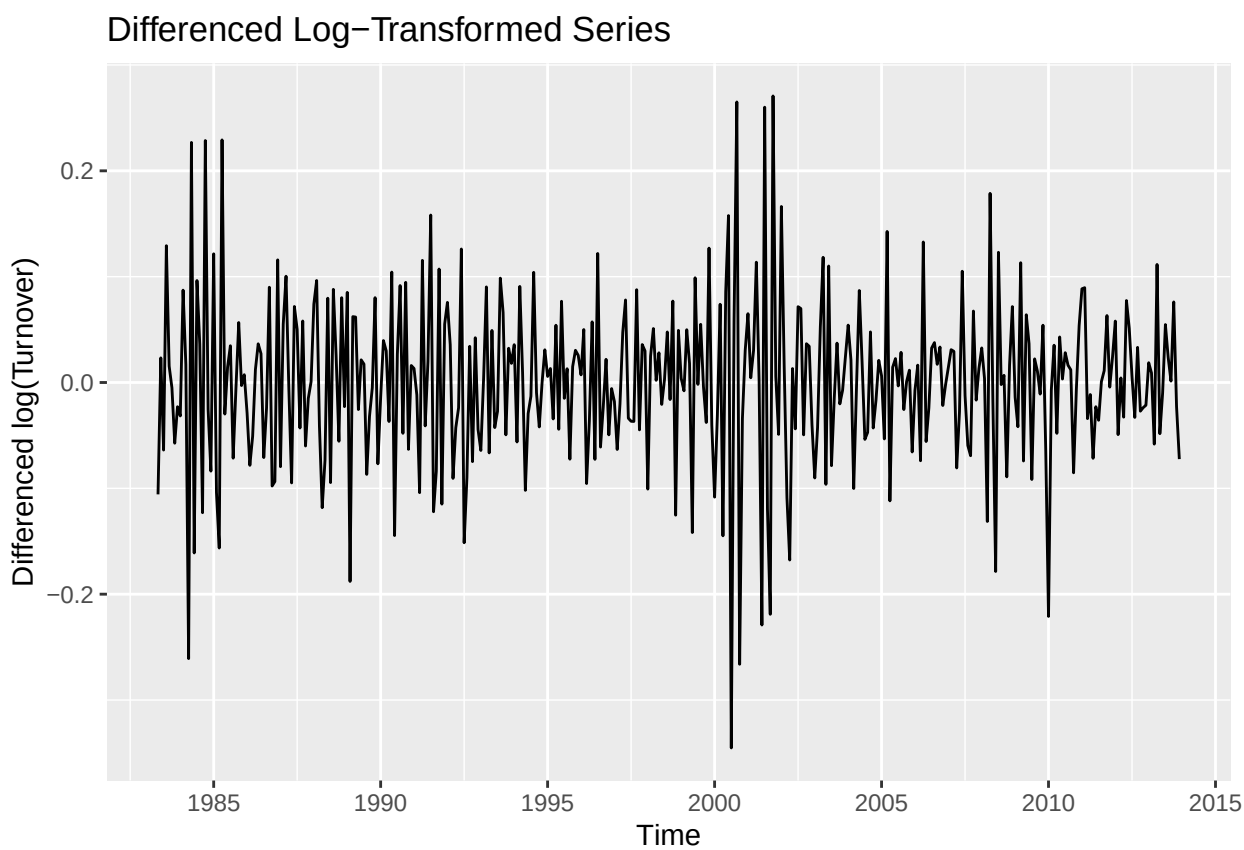
```
## [1] 1
```

```r
nsdiffs(myts_trans)
```

```
## [1] 1
```

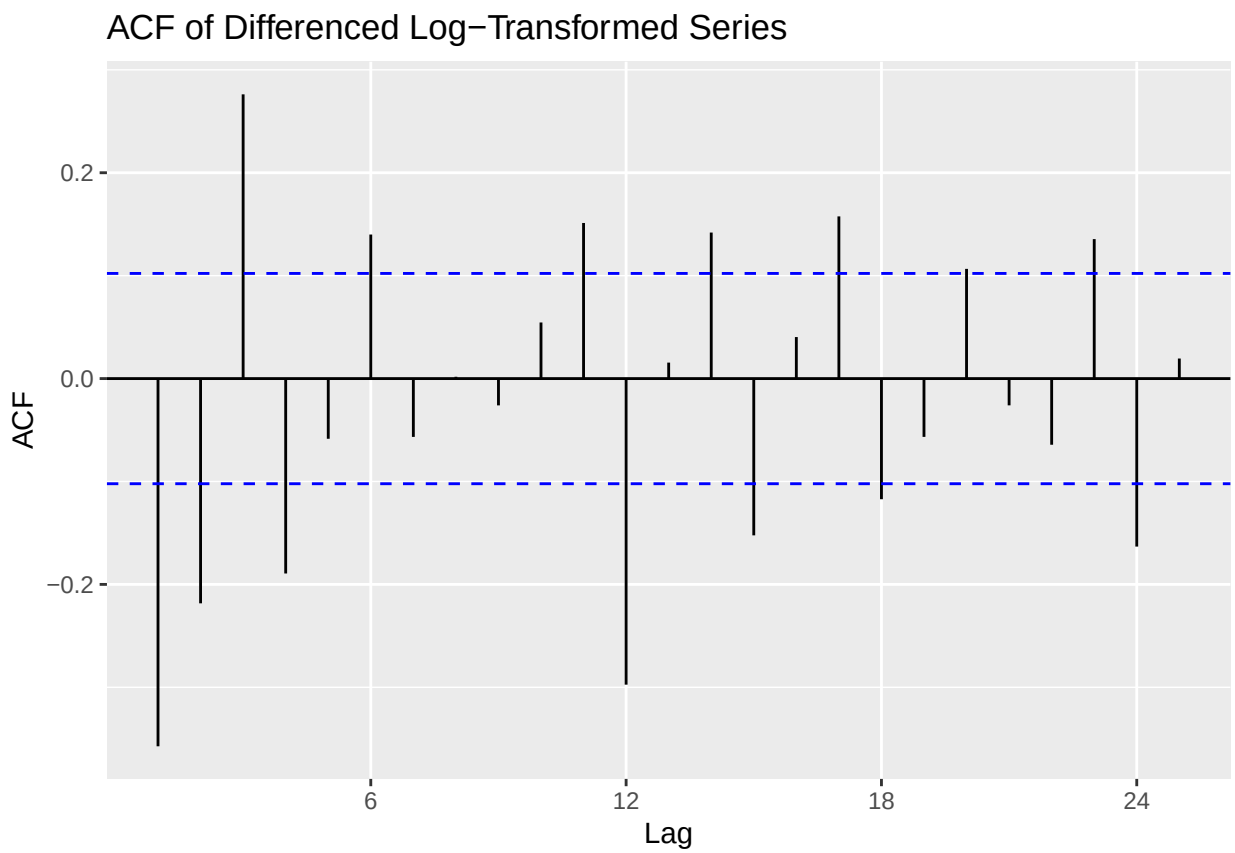
Since ndiffs = 1 and nsdiffs = 1, we need one seasonal difference (lag 12) and one regular difference.

```r
myts_diff <- diff(myts_trans, lag = 12)
myts_diff2 <- diff(myts_diff)

autoplot(myts_diff2) +
   labs(title = "Differenced Log-Transformed Series",
       x = "Time",
       y = "Differenced log(Turnover)")
```



ACF of differenced series

```r
ggAcf(myts_diff2) +
  labs(title = "ACF of Differenced Log-Transformed Series")
```



The transformed series still shows both trend and seasonality, so we need difference to make it stationary. Since the data is monthly but with a strong yearly seasonal pattern, one seasonal difference with lag 12 makes sense to remove the upward trend.

After applying both one seasonal and one regular difference, the series fluctuates around a roughly constant mean. Also, now there is no strong seasonal pattern anymore. So, the appropriate differencing order is one regular difference and one seasonal difference at lag 12.

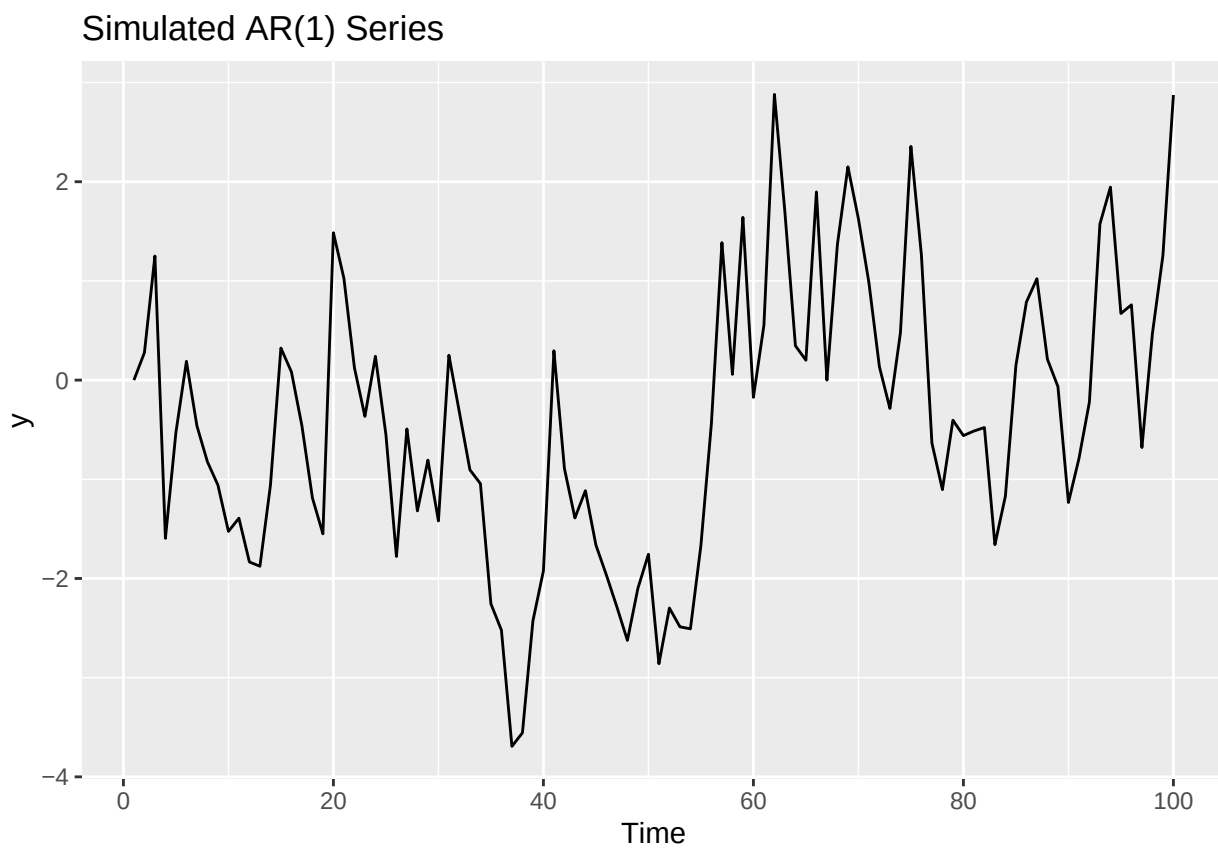
Question 6

Use R to simulate and plot some data from simple ARIMA models.

(a) Use the following R code to generate data from an AR(1) model with
1=0.6 and 2=1. The process starts with y1=0.

```
set.seed(1234)

y <- ts(numeric(100))
e <- rnorm(100)

for(i in 2:100)
  y[i] <- 0.6*y[i-1] + e[i]

autoplot(y) +
  labs(
```

```
    title = "Simulated AR(1) Series",
    x = "Time",
    y = "y"
  )
```

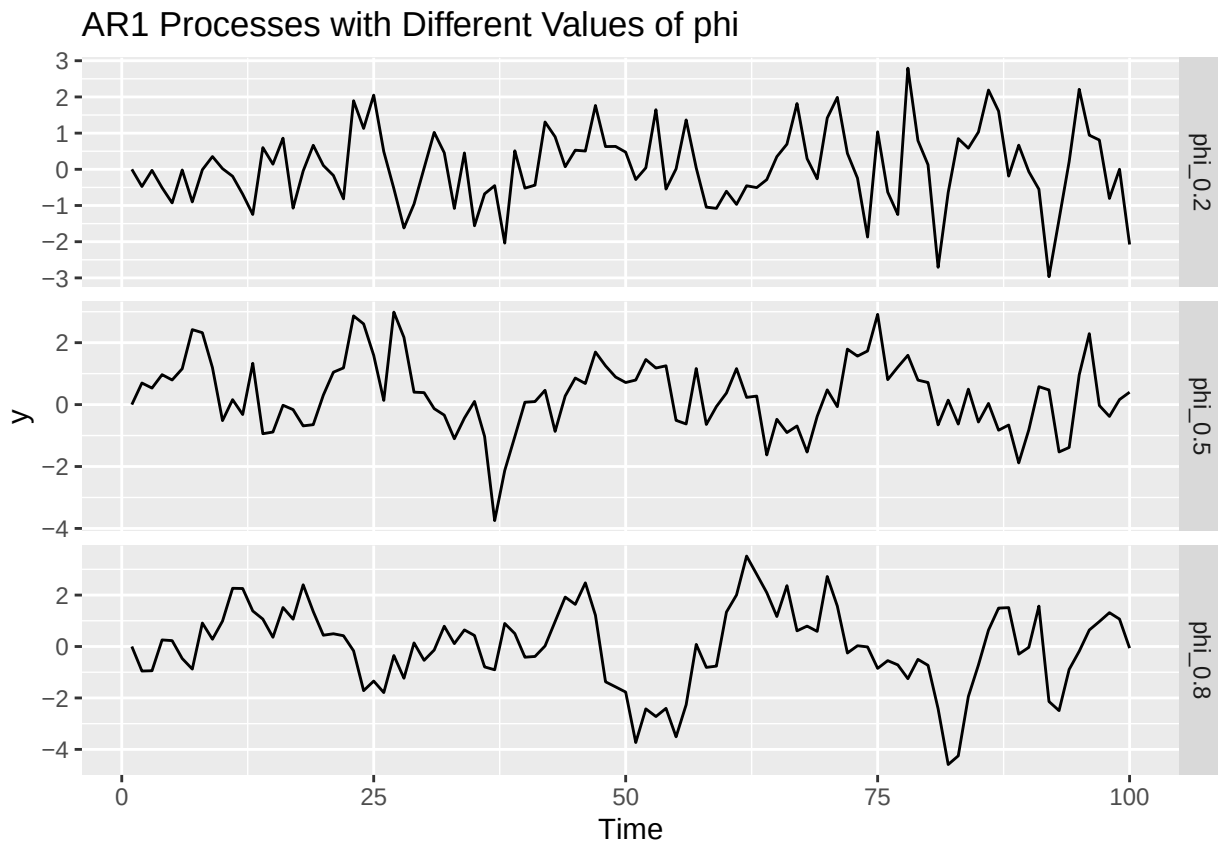


The simulated series shows persistence over time, meaning that observations tend to be related to their recent past values. Because _{1}=0.6, the series tends to move smoothly and values stay above or below zero for short periods before changing direction.

(b) Produce a time plot for the series. How does the plot change as you change _{1}?

```
simulate_ar1 <- function(phi, n = 100){
  y <- ts(numeric(n))
  e <- rnorm(n)

  for(i in 2:n)
    y[i] <- phi*y[i-1] + e[i]

  y
}

y1 <- simulate_ar1(0.2)
y2 <- simulate_ar1(0.5)
y3 <- simulate_ar1(0.8)

autoplot(cbind(phi_0.2 = y1,
```

```
                phi_0.5 = y2,
                phi_0.8 = y3),
          facets = TRUE) +
  labs(title = "AR1 Processes with Different Values of phi",
       x = "Time",
       y = "y")
```



By looking at the output plots for different , we can see that when $_1$ is small (0.2), the series seems like white noise, it looks random since it is constantly changing direction. As $_1$ increases, the series is smoother and with less changing in direction, meaning that values tend to remain above or below the mean for longer periods. For $_1$=0.8, the series has long swings (trends) and a strong dependence on past values.

(c) Write your own code to generate data from an MA(1) model with $\theta_1 = 0.6$ and $\sigma^2 = 1$.

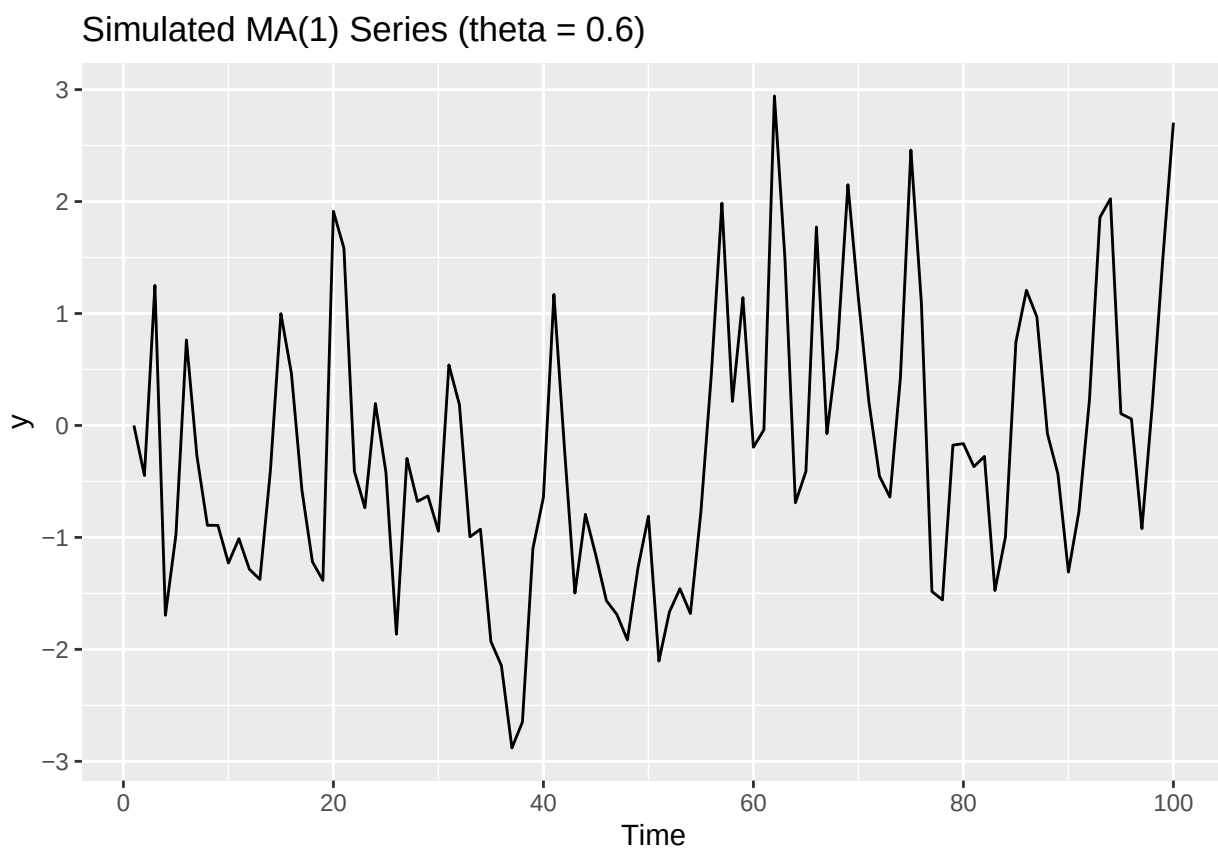
MA(1) model: $y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$ where $\theta_1 = 0.6$ and $\sigma^2 = 1$

```
set.seed(1234)

y_ma1 <- ts(numeric(100))
e <- rnorm(100)

for(i in 2:100){
  y_ma1[i] <- e[i] + 0.6 * e[i-1]
}

head(y_ma1, 10)
```

```
## Time Series:
## Start = 1
## End = 10
## Frequency = 1
##  [1]  0.0000000 -0.4468102  1.2508987 -1.6950330 -0.9782939  0.7635307
##  [7] -0.2711064 -0.8914758 -0.8924311 -1.2287090
```

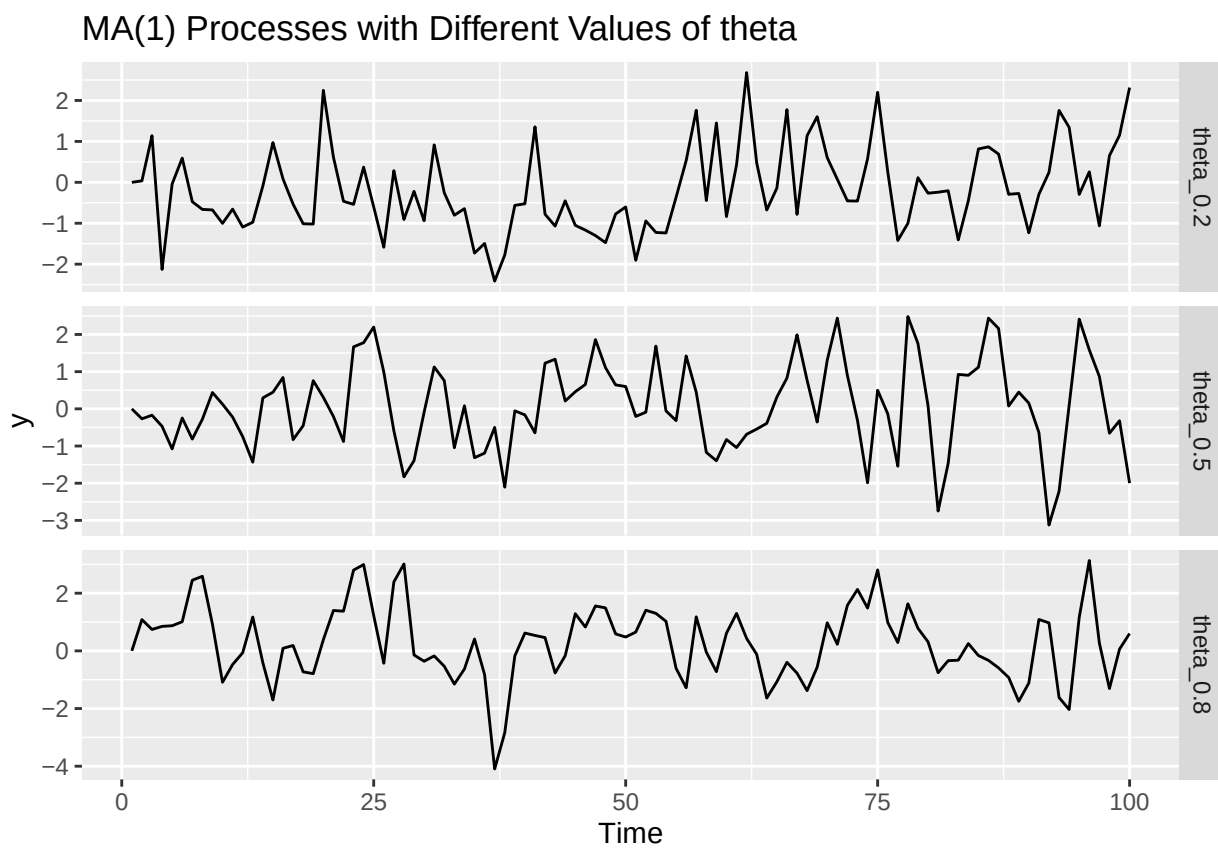
(d) Produce a time plot for the series. How does the plot change as you change $_1$?

```r
autoplot(y_ma1) +
  labs(title = "Simulated MA(1) Series (theta = 0.6)",
       x = "Time",
       y = "y")
```




```r
simulate_ma1 <- function(theta, n = 100){
  y <- ts(numeric(n))
  e <- rnorm(n)

  for(i in 2:n){
    y[i] <- e[i] + theta * e[i-1]
  }

  y
}
```

```r
set.seed(1234)

y1 <- simulate_ma1(0.2)
y2 <- simulate_ma1(0.5)
y3 <- simulate_ma1(0.8)

autoplot(cbind(theta_0.2 = y1,
               theta_0.5 = y2,
               theta_0.8 = y3),
         facets = TRUE) +
  labs(title = "MA(1) Processes with Different Values of theta",
       x = "Time",
       y = "y")
```

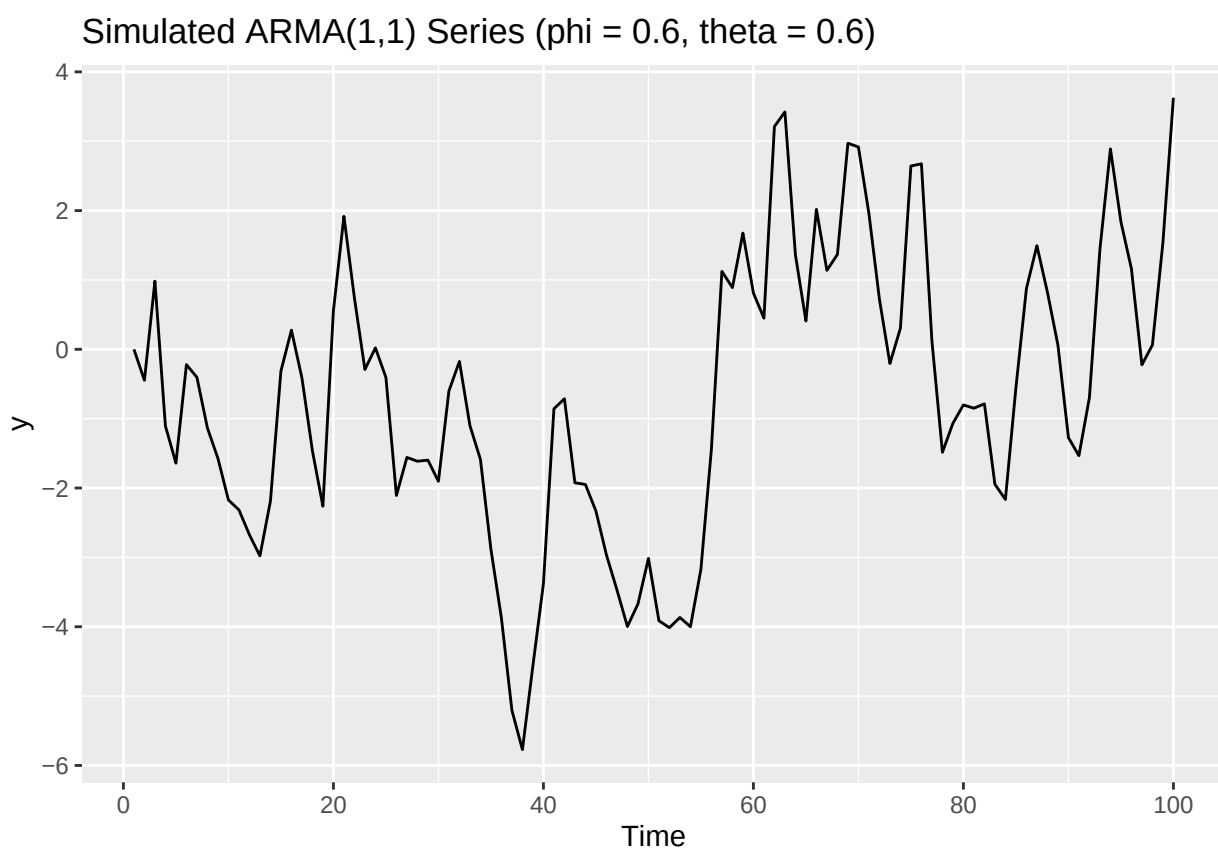


When $_1$ is small, the series behaves similarly to white noise with little a clear dependence between observations. As $_1$ increases, neighboring observations are more related to each other (short memory). However, unlike in the AR(1) process, the effect of extremes does not persist for long periods, that's why the series still changes direction frequently.

(e) Generate data from an ARMA(1,1) model with $\phi_1 = 0.6$, $\theta_1 = 0.6$ and $\sigma_2 = 1$.

ARMA(1,1) model: $y_t = \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$

```r
set.seed(1234)

y_arma <- ts(numeric(100))
e <- rnorm(100)

for(i in 2:100){
  y_arma[i] <- 0.6*y_arma[i-1] + e[i] + 0.6*e[i-1]
}

# time plot
autoplot(y_arma) +
  labs(title = "Simulated ARMA(1,1) Series (phi = 0.6, theta = 0.6)",
       x = "Time",
       y = "y")
```

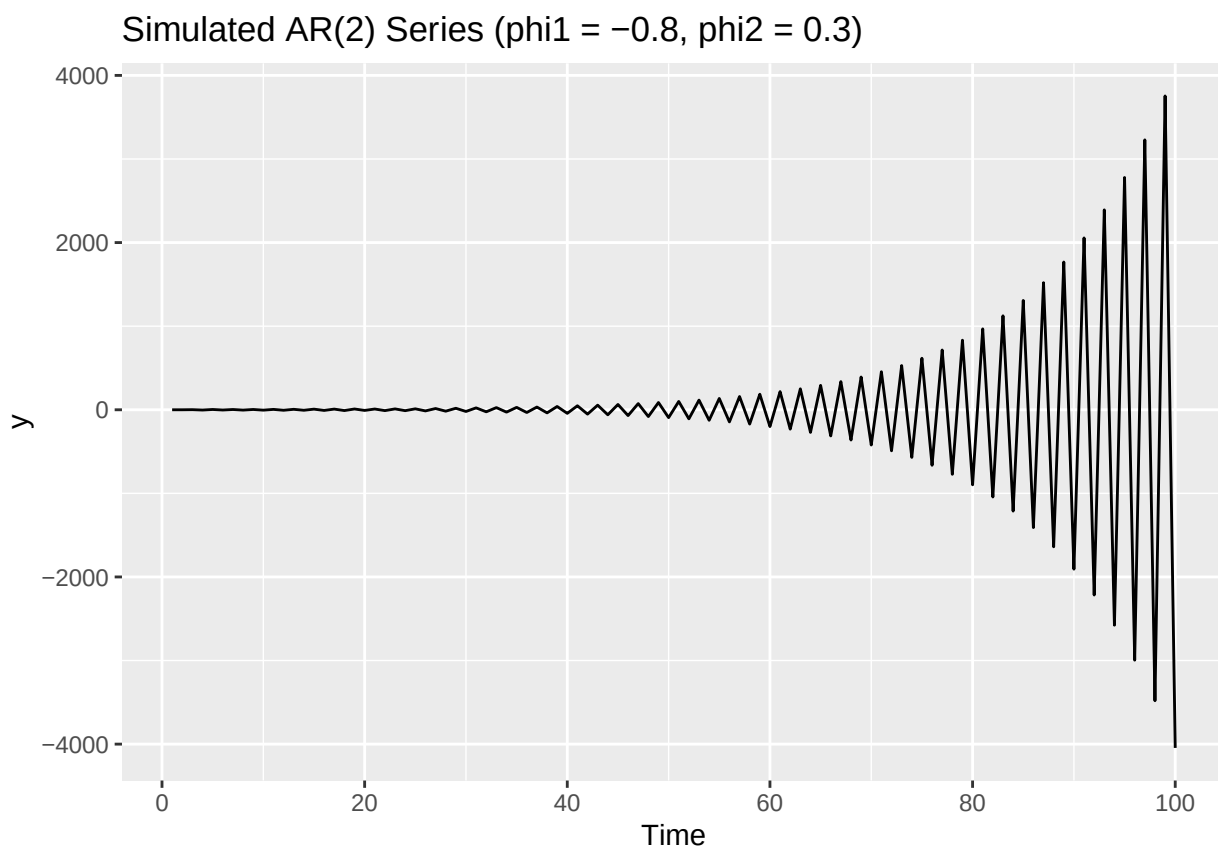


The ARMA(1,1) series combines characteristics of both AR and MA processes. The autoregressive component makes the series more persistence, while the MA component adds short-term effects from recent shocks/behavior. So, the series seems smoother than white noise but still has clear fluctuations over time.

(f) Generate data from an AR(2) model with $\phi_1 = -0.8$, $\phi_2 = 0.3$ and $\sigma_2 = 1$. (Note that these parameters will give a non-stationary series.)

AR(2) model: $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$ with $\phi_1 = -0.8$, $\phi_2 = 0.3$ and $\sigma_2 = 1$

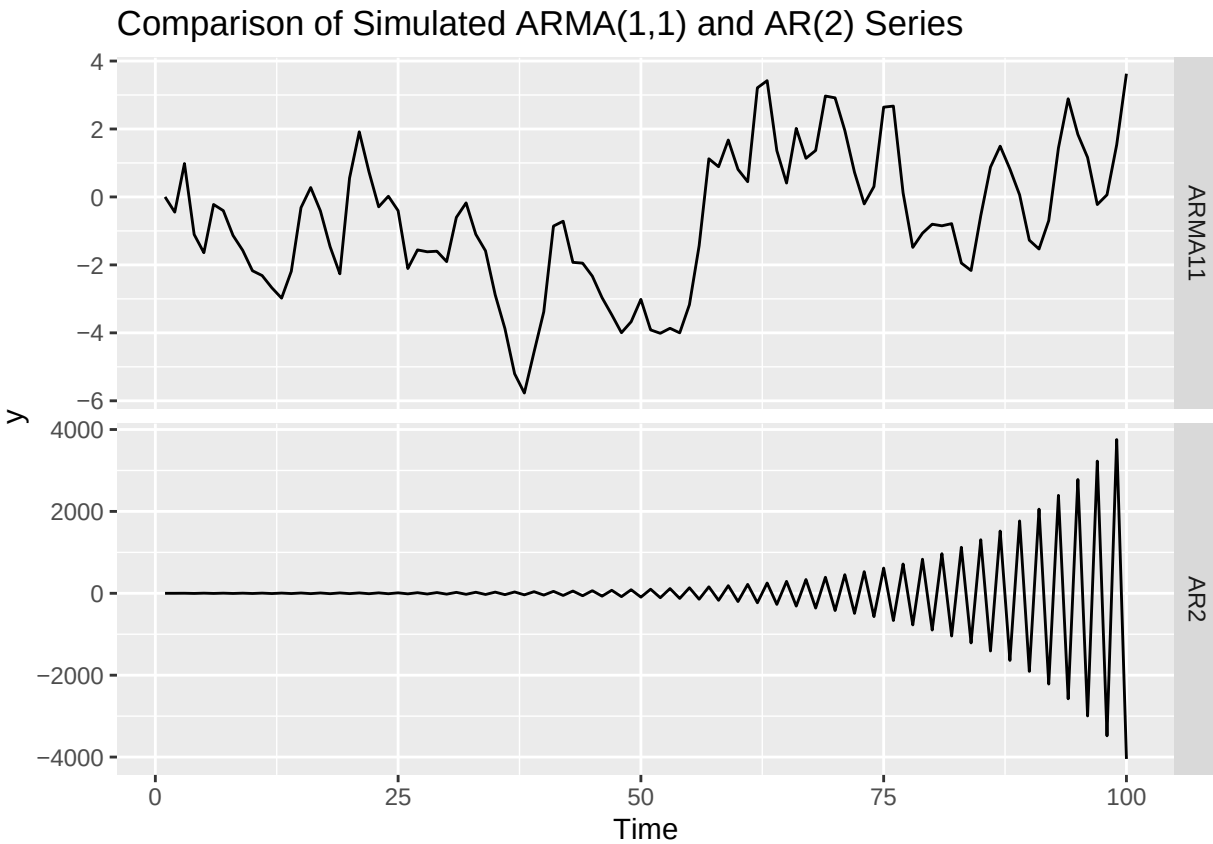
```r
set.seed(1234)

y_ar2 <- ts(numeric(100))
e <- rnorm(100)

for(i in 3:100){
  y_ar2[i] <- -0.8*y_ar2[i-1] + 0.3*y_ar2[i-2] + e[i]
}

# time plot
autoplot(y_ar2) +
  labs(title = "Simulated AR(2) Series (phi1 = -0.8, phi2 = 0.3)",
       x = "Time",
       y = "y")
```



The simulated AR(2) series has an oscillating pattern where the values repeatedly switch between positive and negative (with the same amplitude). Over time, the oscillations become larger (more amplitude), causing the series to diverge away from zero, indicating that the process is non-stationary.

(g) Graph the latter two series and compare them.

```r
autoplot(cbind(ARMA11 = y_arma, AR2 = y_ar2), facets = TRUE) +
  labs(title = "Comparison of Simulated ARMA(1,1) and AR(2) Series",
       x = "Time",
       y = "y")
```



The two series have pretty difference behavior. The ARMA(1,1) series fluctuates around zero and remains relatively stable over time. The size of the fluctuations is pretty consistent, which is typical of a stationary time series.

In contrast, the AR(2) series shows oscillations that grow larger over time. The values constantly change between positive and negative, but the size of these swings increases with time. This causes the series to diverge away from zero, creating the growing behavior seen in the plot.

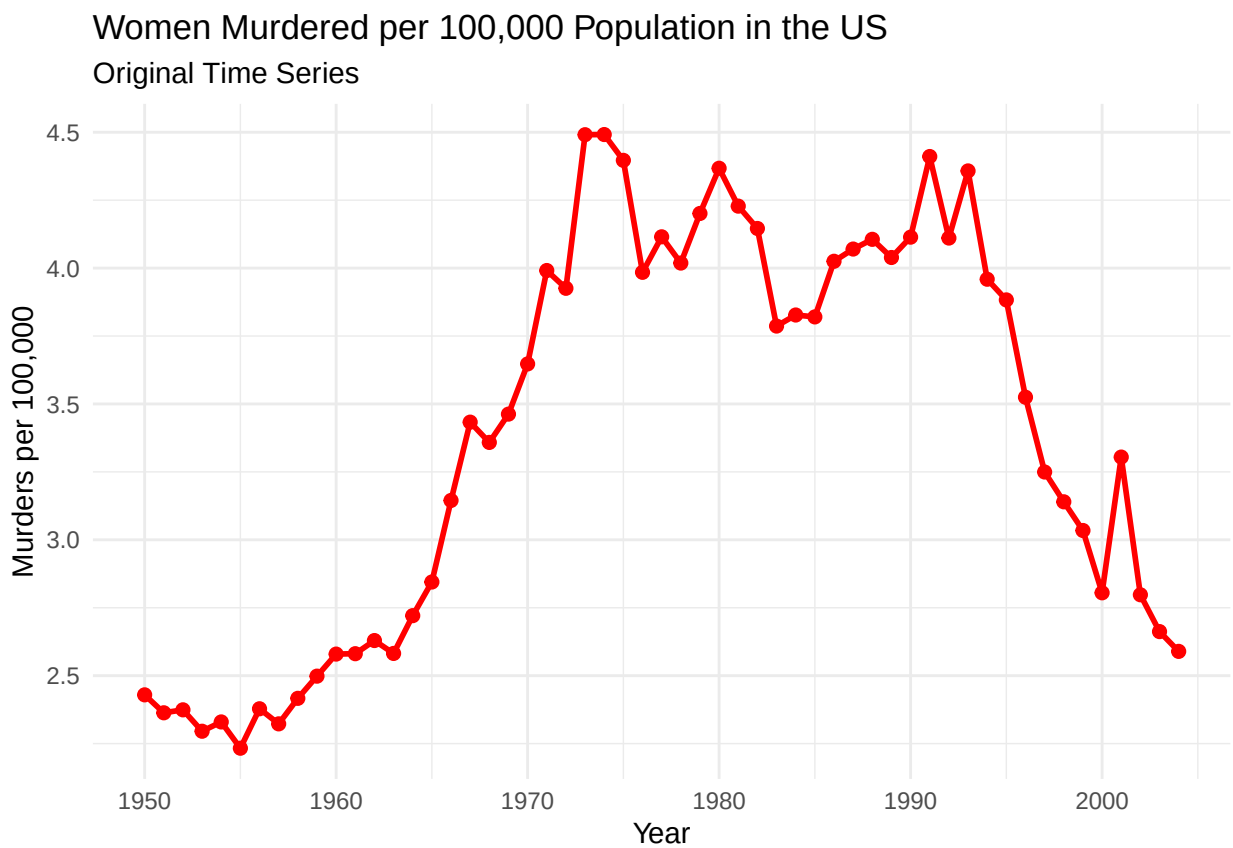
Question 7

Consider wmurders, the number of women murdered each year (per 100,000 standard population) in the United States.

(a) By studying appropriate graphs of the series in R, find an appropriate ARIMA(p,d,q) model for these data.

```
data(wmurders)
```

```
autoplot(wmurders) +
  labs(title = "Women Murdered per 100,000 Population in the US",
       subtitle = "Original Time Series",
       x = "Year",
       y = "Murders per 100,000") +
  theme_minimal() +
  geom_point(color = "red", size = 2) +
  geom_line(color = "red", linewidth = 1)
```



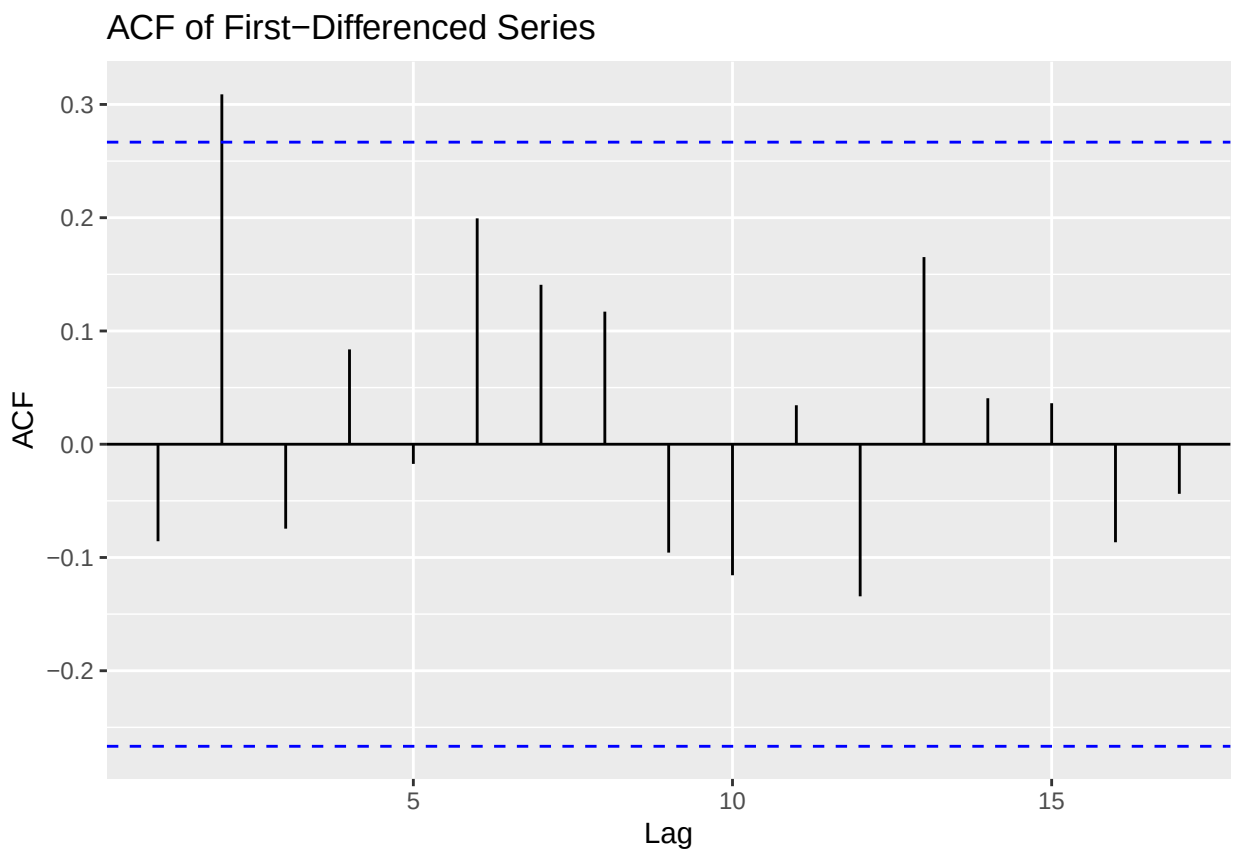
The time series does not look stationary since the mean changes over time. The rate of women murdered rises from mid 1950s into the mid 1970s, stays relatively high through the 1980s and early 1990s, and then declines noticeably toward the end of the series. Since the series behavior changes over time rather than fluctuating around a constant mean, differencing is needed before fitting an ARIMA model.

```
# first difference
wmurders_diff <- diff(wmurders)

autoplot(wmurders_diff) +
  labs(title = "First Differenced Women Murder Series",
       x = "Year",
       y = "Differenced rate")
```

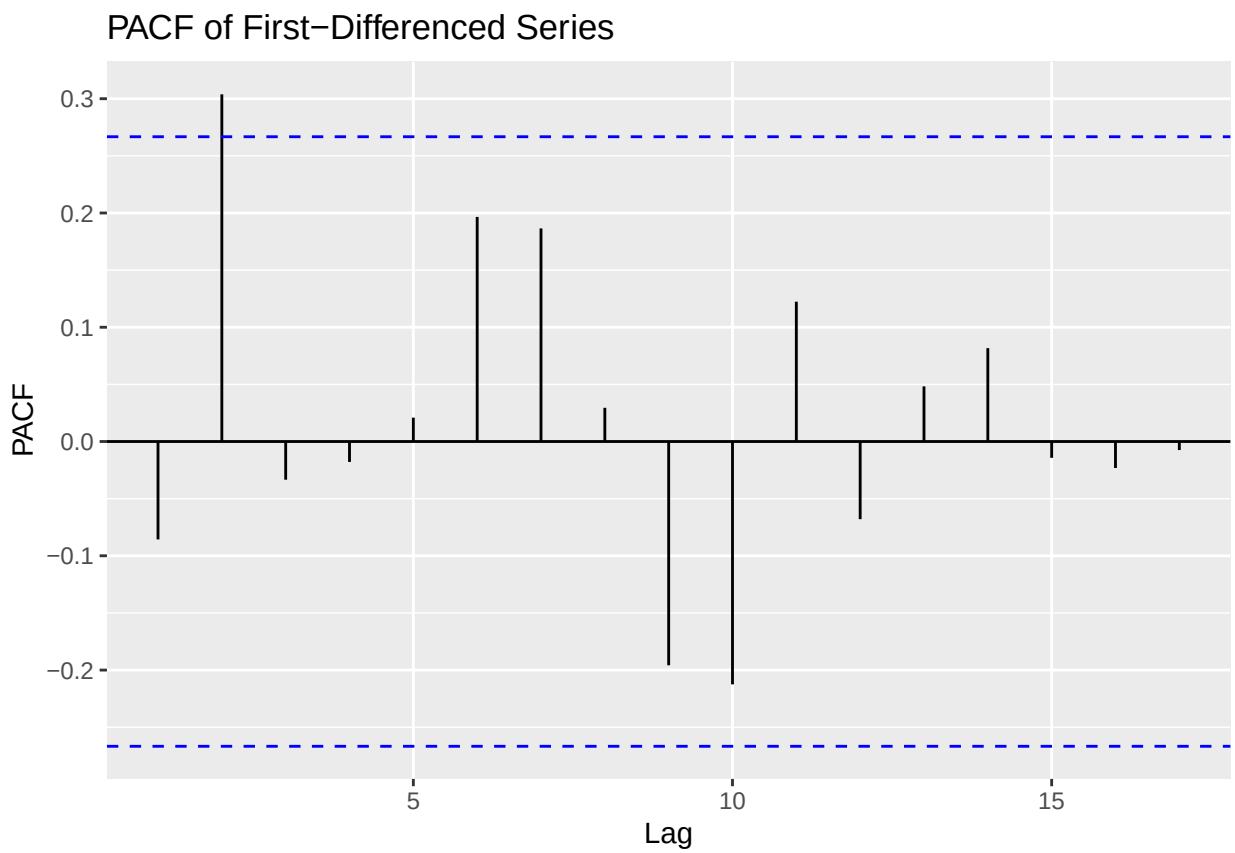
The first differenced series fluctuates around a more stable mean now making first difference appropriate.

Checking **ACF and PACF**

```
ggAcf(wmurders_diff) +
  labs(title = "ACF of First-Differenced Series")
```




```r
ggPacf(wmurders_diff) +
  labs(title = "PACF of First-Differenced Series")
```



(a)

The original series is not stationary because its mean changes over time. The changing level explained above suggests that differencing is required. After taking the first difference, the series fluctuates around a roughly constant mean with no clear trend, so one difference is sufficient (even though the ndiffs(wmurders) gave me 2).

The ACF of the differenced series shows a notable spike at lag 2 while the other lags fall mostly within the confidence bounds. The PACF also shows a significant spike at lag 2. This pattern suggests an autoregressive structure of order 2 after differencing. Therefore, an appropriate model for the series is:

ARIMA(2,1,0)

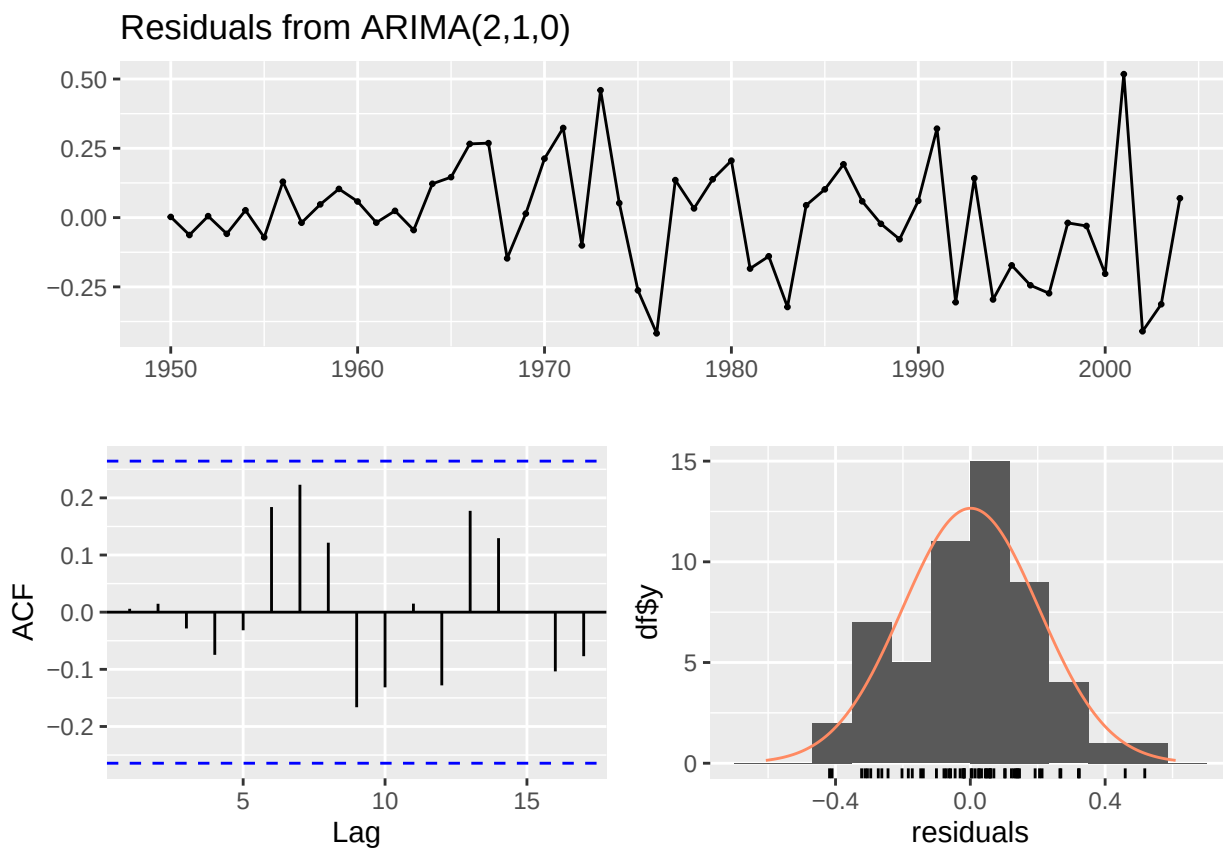
Confirming that the model makes sense

```
Arima(wmurders, order = c(2,1,0))
```

```
## Series: wmurders
## ARIMA(2,1,0)
##
## Coefficients:
##           ar1     ar2
##       -0.0572  0.2967
## s.e.   0.1277  0.1275
##
## sigma^2 = 0.04265:  log likelihood = 9.48
## AIC=-12.96   AICc=-12.48   BIC=-6.99
```

Checking if residuals behave like white noise

```
checkresiduals(Arima(wmurders, order = c(2,1,0)))
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,1,0)
## Q* = 9.9569, df = 8, p-value = 0.2681
##
## Model df: 2.   Total lags used: 10
```

It is confirmed that the residuals behave like white noise.

(b) Should you include a constant in the model? Explain.

Because the model includes one difference (d = 1), adding a constant would imply a linear trend in the original series. The time plot of the data does not show a linear trend over the entire period, so including a constant is not appropriate. Therefore, the ARIMA(2,1,0) model should be fitted without a constant.

(c) Write this model in terms of the backshift operator.

Since p=2, d=1, and q=0

My ARIMA model is: $\phi(B)(1-B)y_t = \varepsilon_t$

The AR(2) polynomial is $\phi(B) = 1 - \phi_1 B - \phi_2 B^2$

finding $_1$ and $_2$

```
Arima(wmurders, order = c(2,1,0))
```

```
## Series: wmurders
## ARIMA(2,1,0)
##
## Coefficients:
##           ar1     ar2
##       -0.0572  0.2967
## s.e.   0.1277  0.1275
##
## sigma^2 = 0.04265:  log likelihood = 9.48
## AIC=-12.96   AICc=-12.48   BIC=-6.99
```

Since $\phi_1 = -0.0572$ and $\phi_2 = 0.2967$

Then, $\phi(B) = 1 - (-0.0572)B - 0.2967B^2$

If we substitute (B) in the ARIMA model: $(1 + 0.0572B - 0.2967B^2)(1 - B)y_t = \varepsilon_t$

(d) Fit the model using R and examine the residuals. Is the model satisfactory?

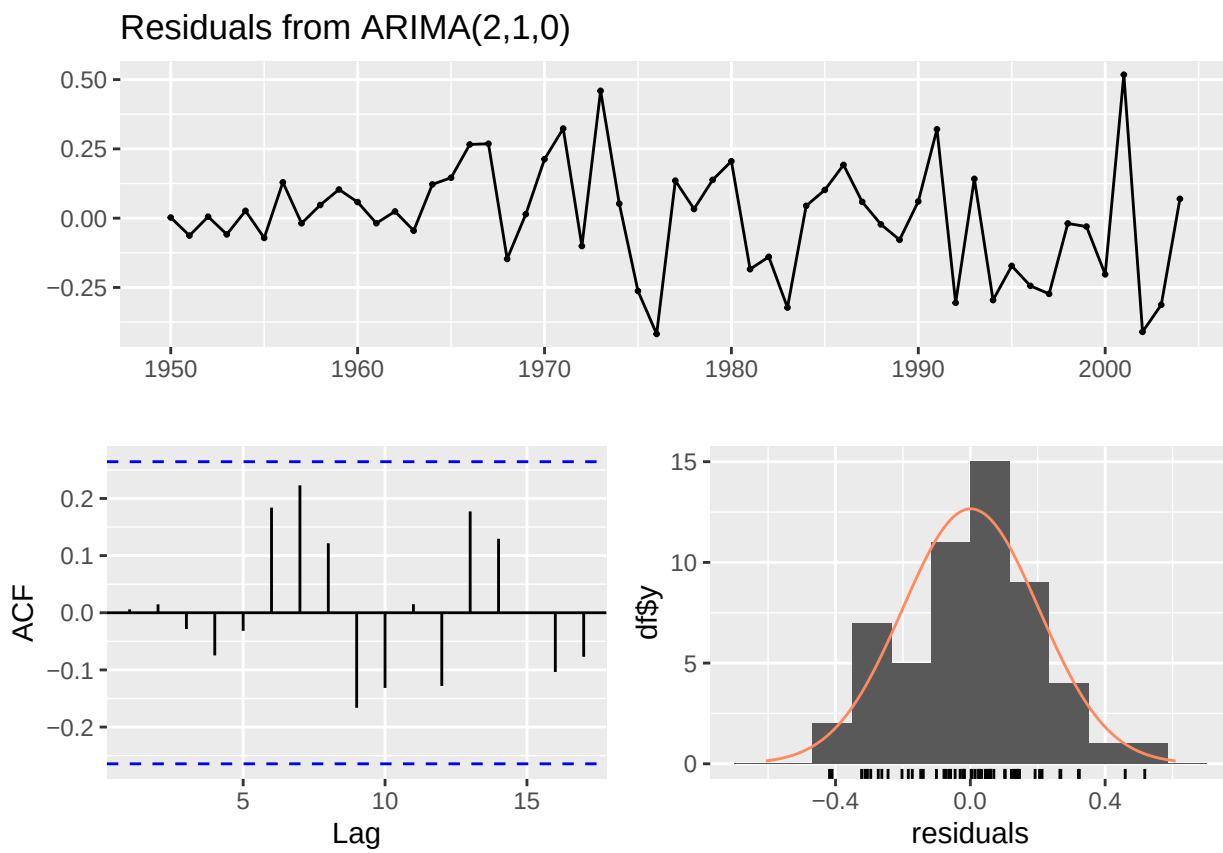
Fit the model (as I did above)

```
fit_wmurders <- Arima(wmurders, order = c(2,1,0))
fit_wmurders
```

```
## Series: wmurders
## ARIMA(2,1,0)
##
## Coefficients:
##           ar1     ar2
##       -0.0572  0.2967
## s.e.   0.1277  0.1275
##
## sigma^2 = 0.04265:  log likelihood = 9.48
## AIC=-12.96   AICc=-12.48   BIC=-6.99
```

Check Residuals

```
checkresiduals(fit_wmurders)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,1,0)
## Q* = 9.9569, df = 8, p-value = 0.2681
##
## Model df: 2.   Total lags used: 10
```

The residual time plot shows that the residuals fluctuate randomly around zero with no obvious trend or pattern. The residual ACF shows that autocorrelations lie within the significance bounds. Also, the histogram of the residuals is approximately symmetric and bell-shaped, suggesting that the residuals are roughly normally distributed.

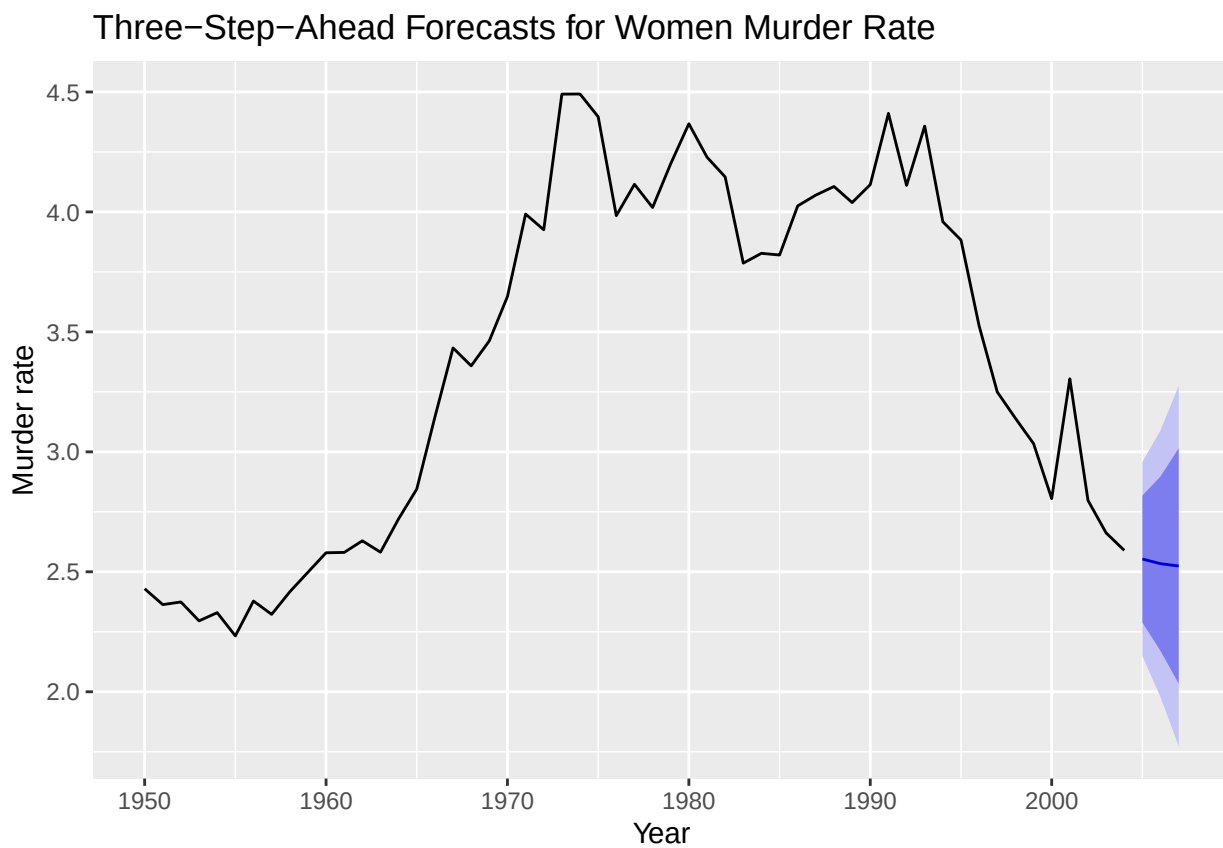
Overall, these results suggest that the residuals behave like white noise, indicating that the ARIMA(2,1,0) model gives a satisfactory fit to the data.

(e) Forecast three times ahead. Check your forecasts by hand to make sure that you know how they have been calculated.

```r
fc_wmurders <- forecast(fit_wmurders, h = 3)
fc_wmurders
```

```
##      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 2005       2.553355 2.288694 2.818015 2.148591 2.958118
## 2006       2.533802 2.170056 2.897548 1.977501 3.090104
## 2007       2.524231 2.033825 3.014637 1.774220 3.274242
```

```r
autoplot(fc_wmurders) +
  labs(title = "Three-Step-Ahead Forecasts for Women Murder Rate",
       x = "Year",
       y = "Murder rate")
```



Checking by hand, which means to

```r
# last observed values
yT   <- tail(wmurders, 1)
yT1  <- tail(wmurders, 2)[1]
yT2  <- tail(wmurders, 3)[1]

# last two differences
wT   <- yT - yT1
wT1  <- yT1 - yT2

# AR coefficients
phi1 <- coef(fit_wmurders)["ar1"]
phi2 <- coef(fit_wmurders)["ar2"]

# forecasts differenced series
w_hat1 <- phi1 * wT + phi2 * wT1
w_hat2 <- phi1 * w_hat1 + phi2 * wT
w_hat3 <- phi1 * w_hat2 + phi2 * w_hat1

# convert back to original scale
```

```r
y_hat1 <- yT + w_hat1
y_hat2 <- y_hat1 + w_hat2
y_hat3 <- y_hat2 + w_hat3

c(y_hat1, y_hat2, y_hat3)
```

```
##       yT   y_hat1   y_hat2
## 2.553355 2.533802 2.524231
```

```r
fc_wmurders$mean
```

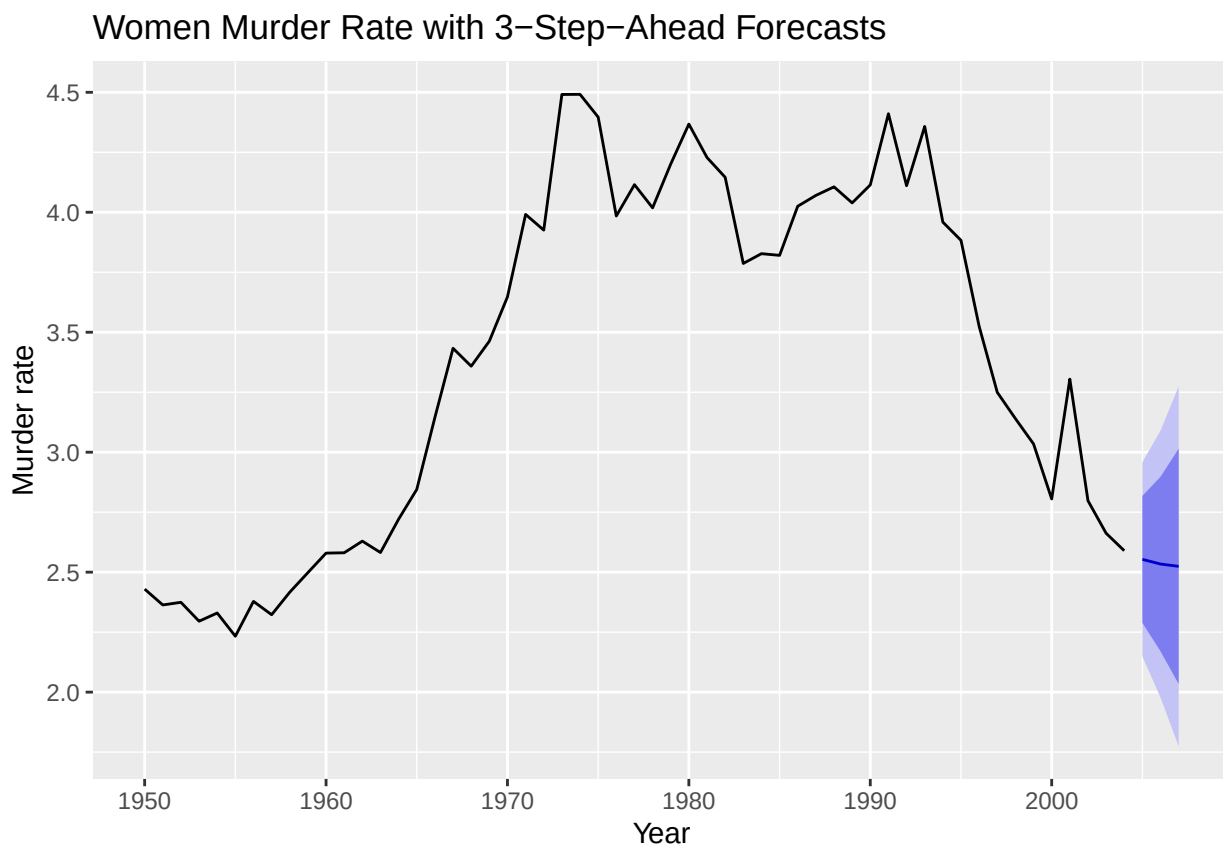
```
## Time Series:
## Start = 2005
## End = 2007
## Frequency = 1
## [1] 2.553355 2.533802 2.524231
```

Using the estimated coefficients ar1 and ar2, each forecast depends on the two most recent differenced values. The 1 step ahead forecast is calculated first, and the two-step and three-step forecasts are then obtained using the previous one using the earlier forecasted differences. Converting these back to the original scale gives the final forecasts for the murder rate.

The hand calculations match the values produced by R, which confirms that the forecasts are accurate.

(f) Create a plot of the series with forecasts and prediction intervals for the next three periods shown.

```r
fc_wmurders <- forecast(fit_wmurders, h = 3)

autoplot(fc_wmurders) +
  labs(title = "Women Murder Rate with 3-Step-Ahead Forecasts",
       x = "Year",
       y = "Murder rate")
```



(g) Does auto.arima() give the same model you have chosen? If not, which model do you think is better?

```
auto.arima(wmurders)
```

```
## Series: wmurders
## ARIMA(1,2,1)
##
## Coefficients:
##           ar1      ma1
##       -0.2434  -0.8261
## s.e.   0.1553   0.1143
##
## sigma^2 = 0.04632:  log likelihood = 6.44
## AIC=-6.88   AICc=-6.39   BIC=-0.97
```

The `auto.arima()` function selected an ARIMA(1,2,1) model, which is different than the ARIMA(2,1,0) I selected. The automatic model uses two differences, which may indicate over-differencing because the first-differenced series already seemed stationary. Since the ARIMA(2,1,0) model produces satisfactory residual diagnostics and aligns with the behavior observed in the ACF and PACF plots, it is the preferred model.

Question 8

Consider austa, the total international visitors to Australia (in millions) for the period 1980-2015.

(a) Use auto.arima() to find an appropriate ARIMA model. What model was selected. Check that the residuals look like white noise. Plot forecasts for the next 10 periods.

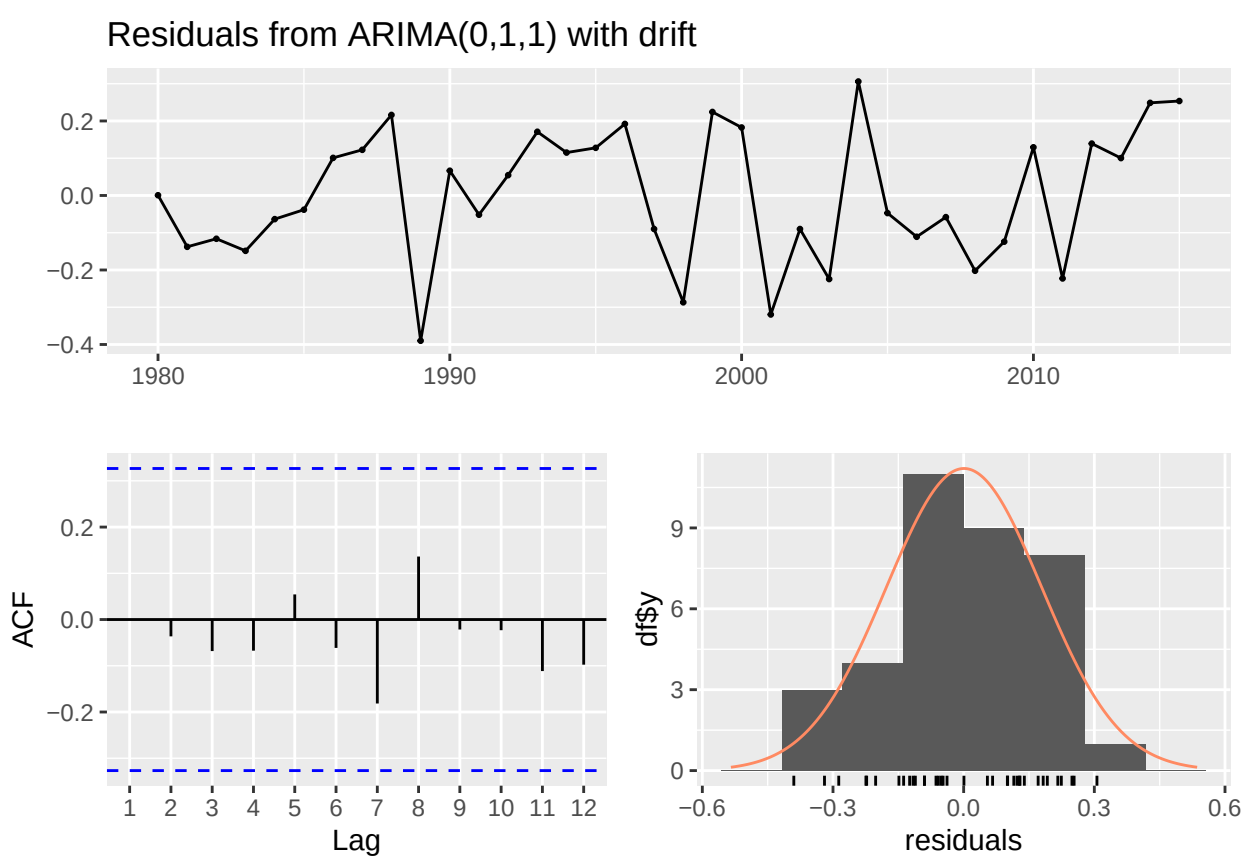
```
fit_austa <- auto.arima(austa)
fit_austa
```

```
## Series: austa
## ARIMA(0,1,1) with drift
##
## Coefficients:
##          ma1   drift
##       0.3006  0.1735
## s.e.  0.1647  0.0390
##
## sigma^2 = 0.03376:  log likelihood = 10.62
## AIC=-15.24   AICc=-14.46   BIC=-10.57
```

The model selected by auto.arima() is ARIMA(0,1,1) model with drift. The MA(1) coefficient is 0.3006 and the drift term is 0.1735, meaning that there is a gradual upward trend in the number of international visitors to Australia over time.

Checking Residuals

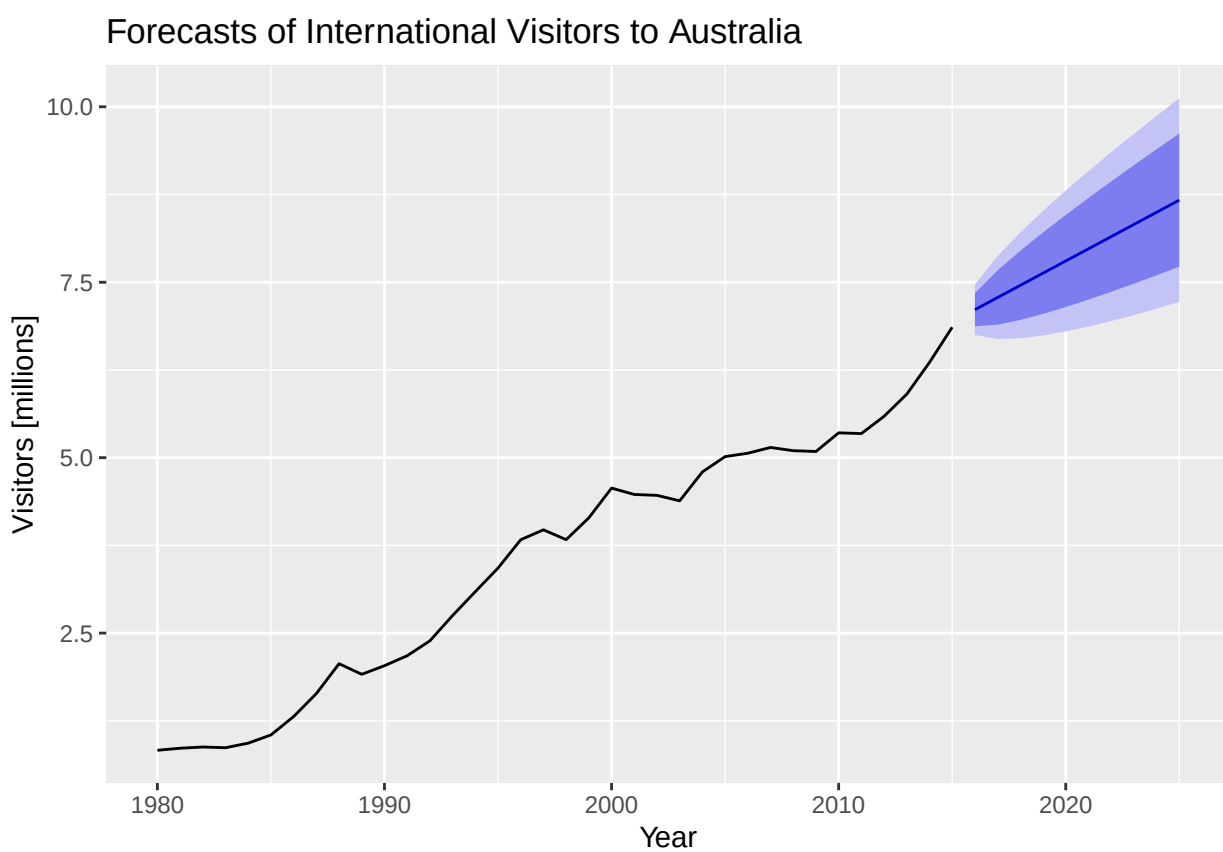
```
checkresiduals(fit_austa)
```



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1) with drift
## Q* = 2.297, df = 6, p-value = 0.8905
##
## Model df: 1.   Total lags used: 7
```

We can confirm that the residuals look like white noise and that there is no significant autocorrelation remaining. So, the model properly captures the structure of the series.

```r
forecast_austa <- forecast(fit_austa, h = 10)

autoplot(forecast_austa) +
  labs(title = "Forecasts of International Visitors to Australia",
       x = "Year",
       y = "Visitors [millions]")
```



Forecasts for the next 10 periods show an upward trend in the number of international visitors to Australia. Naturally, the prediction intervals are larger with time (more uncertainty).

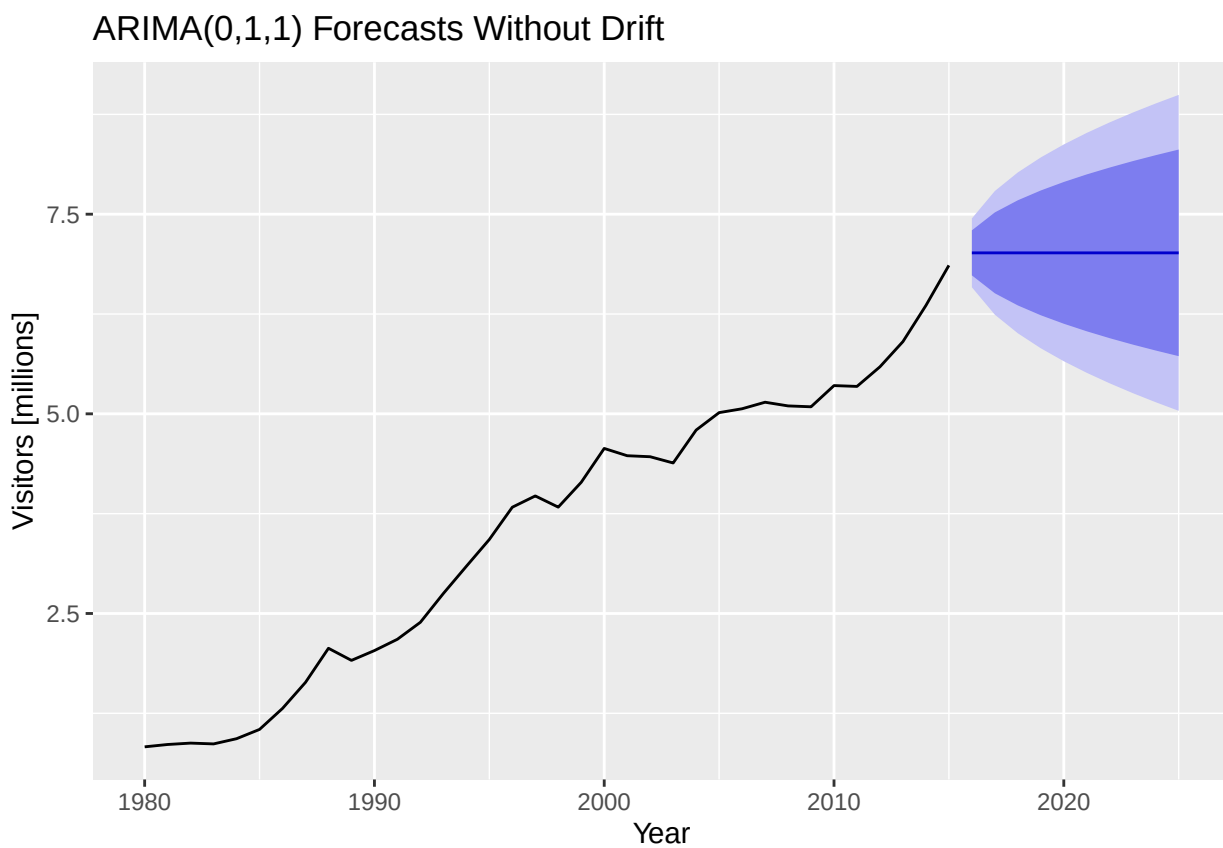
(b) Plot forecasts from an ARIMA(0,1,1) model with no drift and compare these to part a. Remove the MA term and plot again.

ARIMA(0,1,1) with no drift

```r
fit_austa_nodrift <- Arima(austa, order = c(0,1,1), include.drift = FALSE)
fit_austa_nodrift
```

```
## Series: austa
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.4936
## s.e.  0.1265
##
## sigma^2 = 0.04833:  log likelihood = 3.73
## AIC=-3.45   AICc=-3.08   BIC=-0.34
```

```r
# plot with forecast
fc_austa_nodrift <- forecast(fit_austa_nodrift, h = 10)

autoplot(fc_austa_nodrift) +
  labs(title = "ARIMA(0,1,1) Forecasts Without Drift",
       x = "Year",
       y = "Visitors [millions]")
```



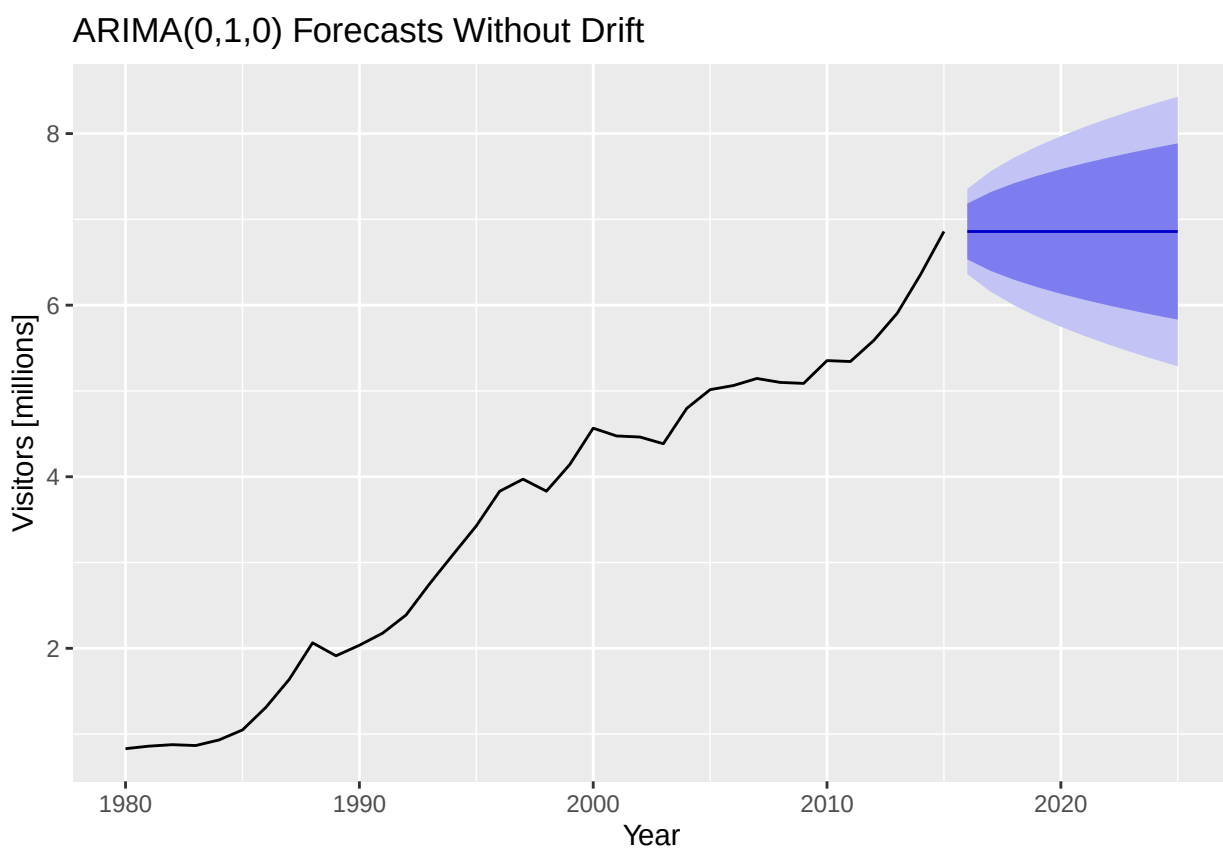
Now the forecasts without drift are flat when compared with (a) because the model no longer includes a trend term. In part (a), the drift term produced forecasts that continued to increase over time. Without drift, the forecasts are more level and do not consider the long-run upward movement as strongly.

If we remove the MA: ARIMA(0,1,0)

```
fit_austa_r <- Arima(austa, order = c(0,1,0), include.drift = FALSE)
fit_austa_r
```

```
## Series: austa
## ARIMA(0,1,0)
##
## sigma^2 = 0.06423:  log likelihood = -1.62
## AIC=5.24   AICc=5.36   BIC=6.8
```

```
# plot with forecasts
fc_austa_r <- forecast(fit_austa_r, h = 10)

autoplot(fc_austa_r) +
  labs(title = "ARIMA(0,1,0) Forecasts Without Drift",
       x = "Year",
       y = "Visitors [millions]")
```



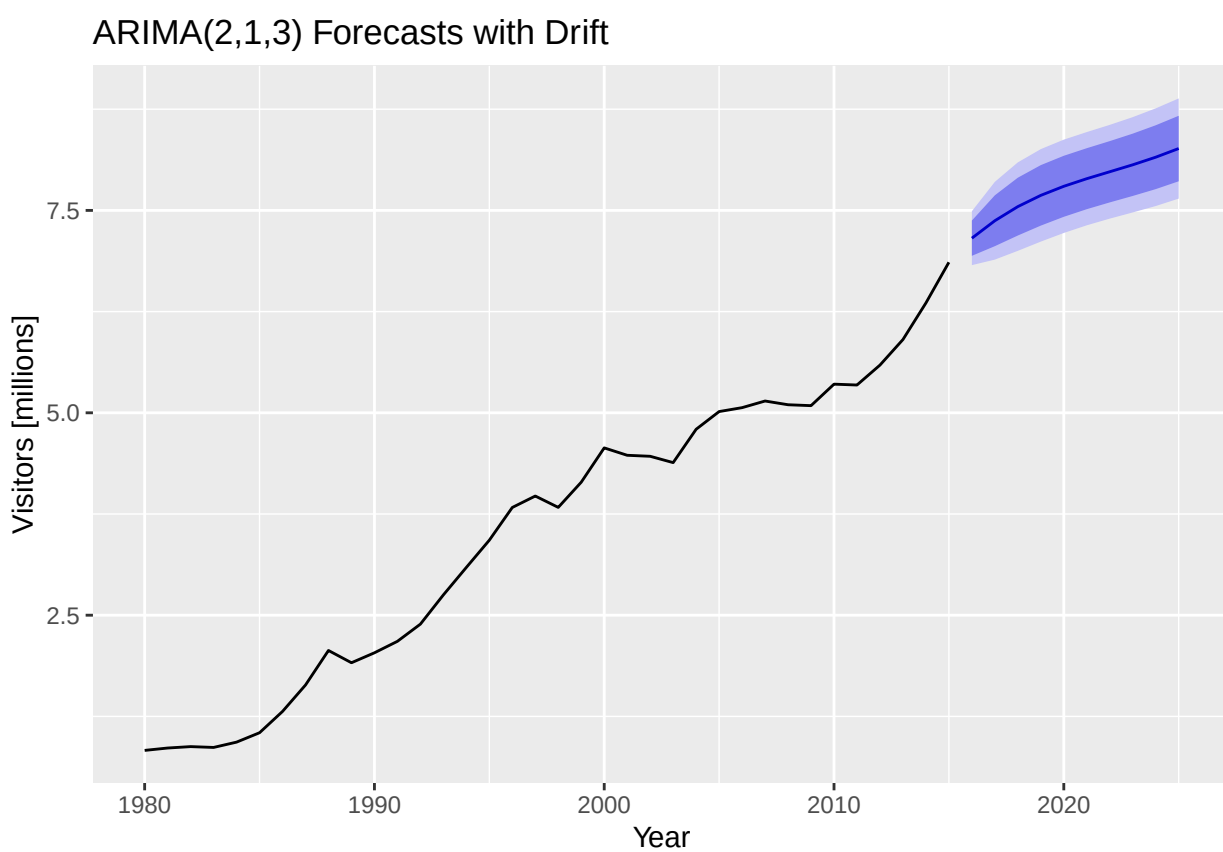
The forecasts from the ARIMA(0,1,1) model without drift and the ARIMA(0,1,0) model seem very similar. In both cases, the forecasts remain approximately flat at the last observed value of the series. This happens because both models use one difference and do not include a drift term. Without drift, the models do not project a trend, so the best forecast for future values is simply the most recent observation. The ARIMA(0,1,1) model accounts for short-term dependence through the MA term, while the ARIMA(0,1,0) model is a simple random walk.

(c) Plot forecasts from an ARIMA(2,1,3) model with drift. Remove the constant and see what happens.

```
fit_austa_213_drift <- Arima(austa, order = c(2,1,3), include.drift = TRUE)
fit_austa_213_drift
```

```
## Series: austa
## ARIMA(2,1,3) with drift
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3   drift
##       1.7648  -0.8513  -1.7684  0.5571  0.2224  0.1725
## s.e.  0.1136   0.1182   0.2433  0.4351  0.2150  0.0051
##
## sigma^2 = 0.0276:  log likelihood = 13.72
## AIC=-13.44   AICc=-9.29   BIC=-2.55
```

```
# plot forecasts
fc_austa_213_drift <- forecast(fit_austa_213_drift, h = 10)

autoplot(fc_austa_213_drift) +
  labs(title = "ARIMA(2,1,3) Forecasts with Drift",
       x = "Year",
       y = "Visitors [millions]")
```

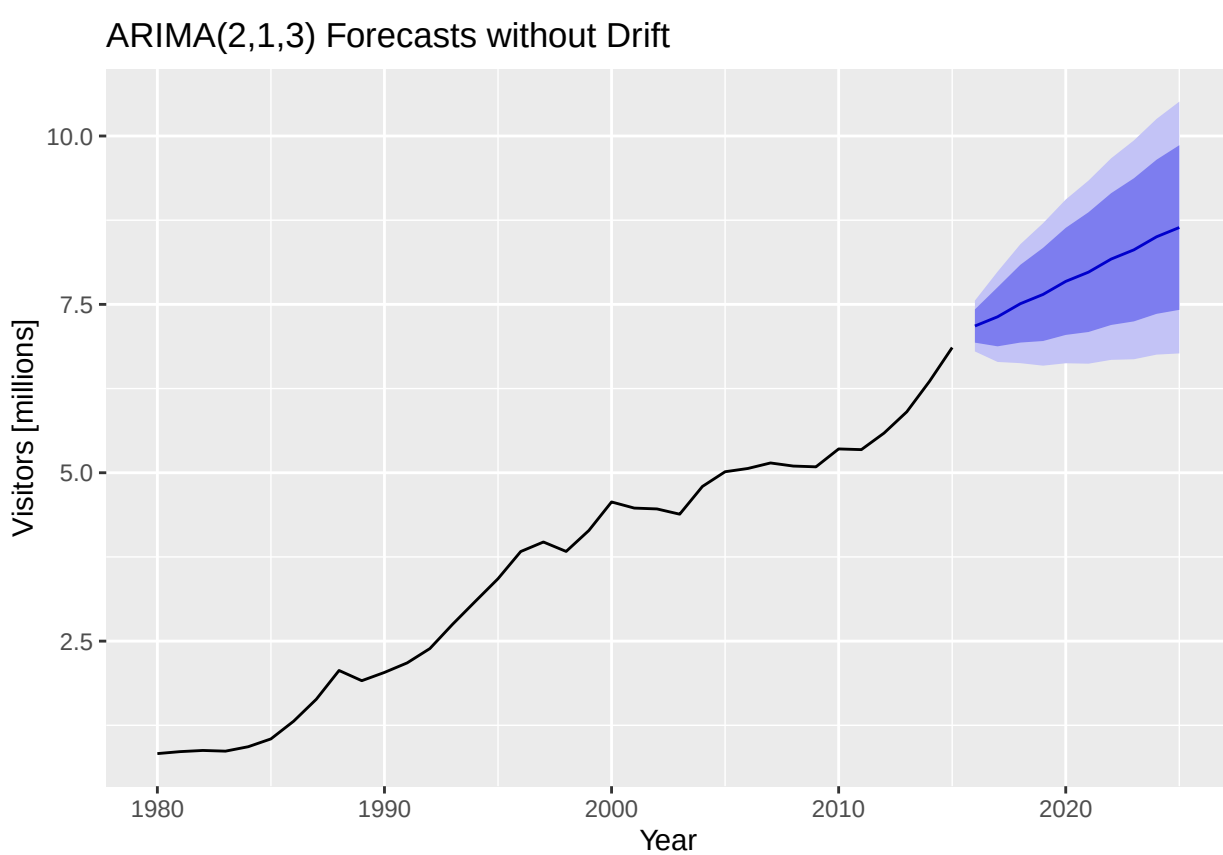


Without constant

```r
fit_austa_213_nodrift <- Arima(austa, order = c(2,1,3),
                               include.drift = FALSE,
                               method = "ML")
fit_austa_213_nodrift
```

```
## Series: austa
## ARIMA(2,1,3)
##
## Coefficients:
##          ar1     ar2     ma1      ma2      ma3
##       0.0004  0.9996  0.4633  -0.9893  -0.4625
## s.e.  0.0031  0.0031  0.2283   0.0329   0.2261
##
## sigma^2 = 0.03568:  log likelihood = 9.24
## AIC=-6.48   AICc=-3.48   BIC=2.85
```

```r
# plot forecasts
fc_austa_213_nodrift <- forecast(fit_austa_213_nodrift, h = 10)

autoplot(fc_austa_213_nodrift) +
  labs(title = "ARIMA(2,1,3) Forecasts without Drift",
       x = "Year",
       y = "Visitors [millions]")
```



The forecasts for the ARIMA(2,1,3) model without drift still show an increasing trend even though the

constant term was removed. This happens because the autoregressive structure, particularly the second AR coefficient which is close to 1, produces strong persistence in the series.

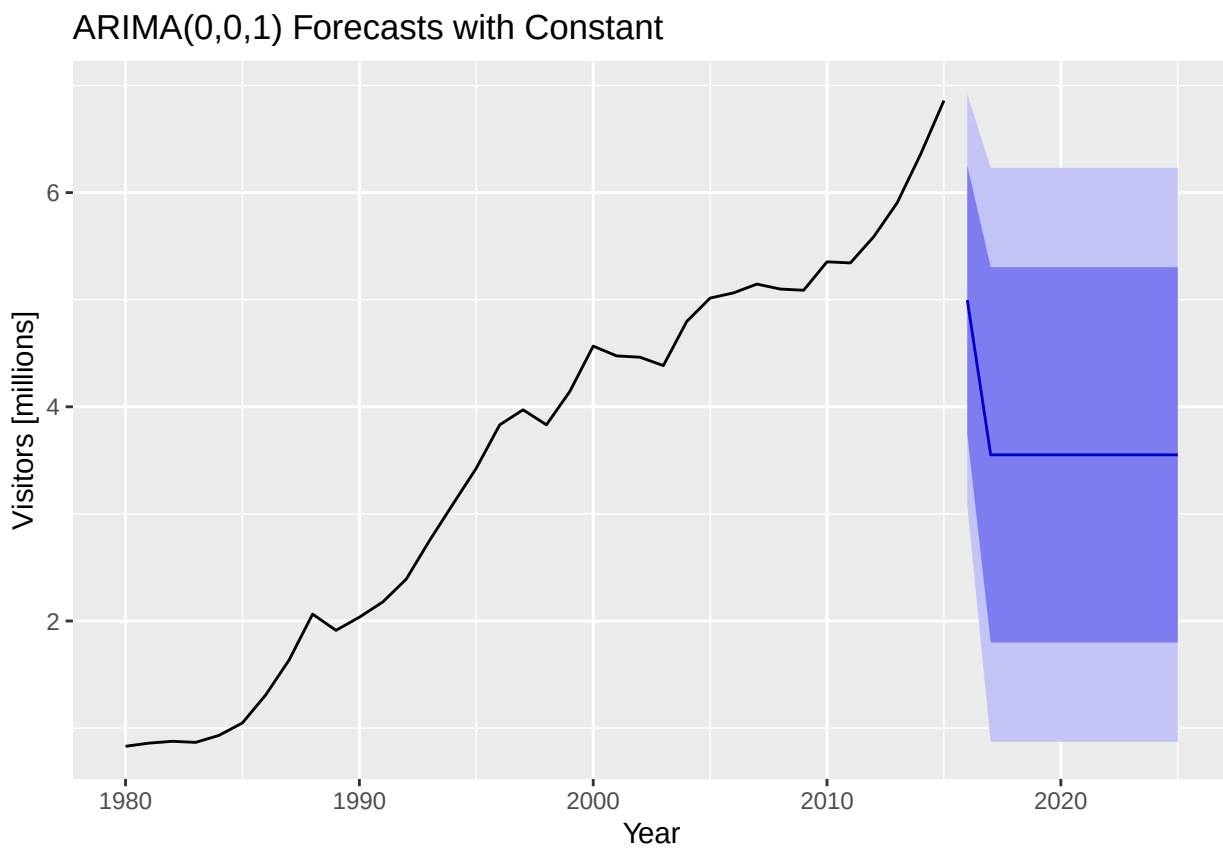
(d) Plot forecasts from an ARIMA(0,0,1) model with a constant. Remove the MA term and plot again.

ARIMA(0,0,1) with a constant

```r
fit_austa_001 <- Arima(austa, order = c(0,0,1), include.mean = TRUE)
fit_austa_001
```

```
## Series: austa
## ARIMA(0,0,1) with non-zero mean
##
## Coefficients:
##          ma1    mean
##       1.0000  3.5515
## s.e.  0.0907  0.3090
##
## sigma^2 = 0.9347:  log likelihood = -50.64
## AIC=107.28   AICc=108.03   BIC=112.04
```

```r
# plot forecasts
fc_austa_001 <- forecast(fit_austa_001, h = 10)

autoplot(fc_austa_001) +
  labs(title = "ARIMA(0,0,1) Forecasts with Constant",
       x = "Year",
       y = "Visitors [millions]")
```

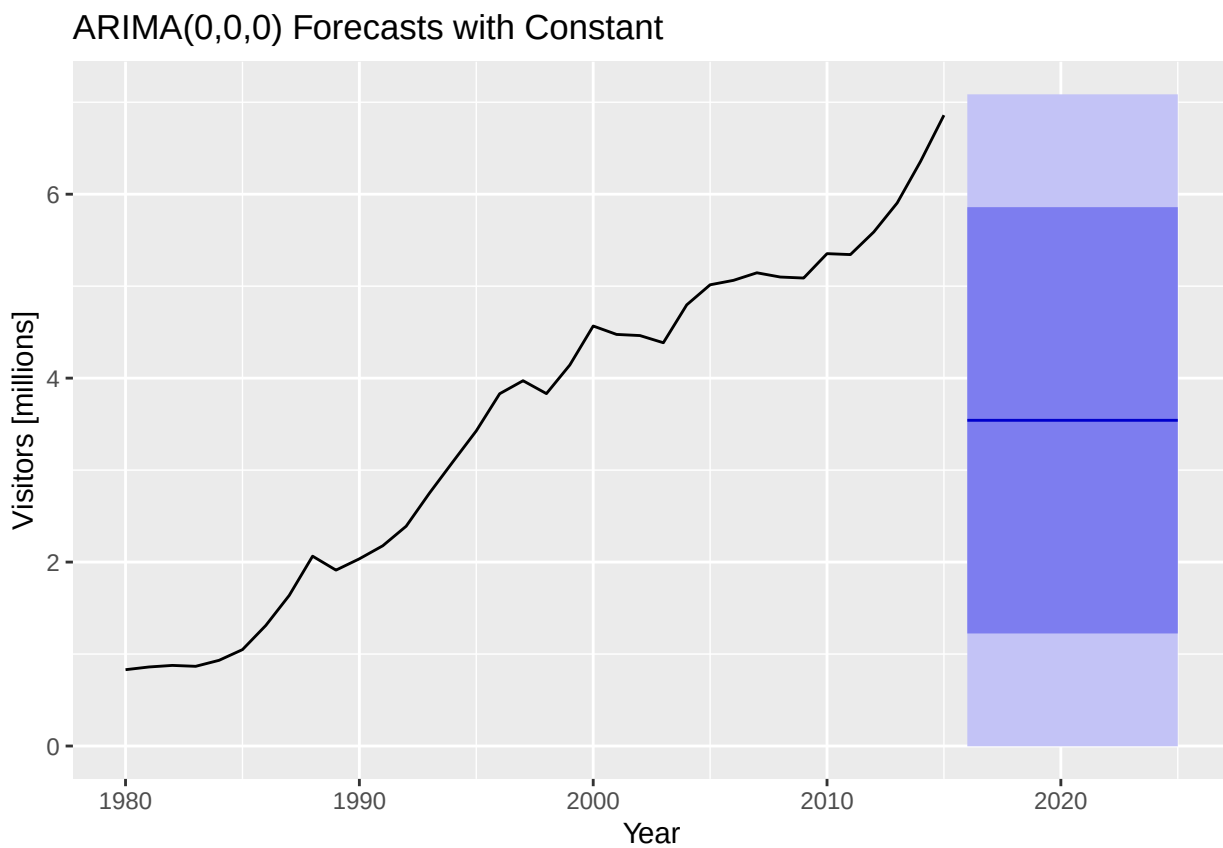


Remove MA: ARIMA(0,0,0)

```
fit_austa_000 <- Arima(austa, order = c(0,0,0), include.mean = TRUE)
fit_austa_000
```

```
## Series: austa
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##         mean
##       3.5414
## s.e.  0.2972
##
## sigma^2 = 3.27:  log likelihood = -71.9
## AIC=147.8   AICc=148.17   BIC=150.97
```

```
# plot forecasts
fc_austa_000 <- forecast(fit_austa_000, h = 10)

autoplot(fc_austa_000) +
  labs(title = "ARIMA(0,0,0) Forecasts with Constant",
       x = "Year",
       y = "Visitors [millions]")
```



The ARIMA(0,0,1) model with a constant produces forecasts that quickly gets close to the estimated mean of about 3.55 million. The MA term allows a short adjustment from the last observed value before the forecasts approaches the mean.

When the MA term is removed, the ARIMA(0,0,0) model simply forecasts the mean for all future periods. Both models ignore the strong upward trend in the data.

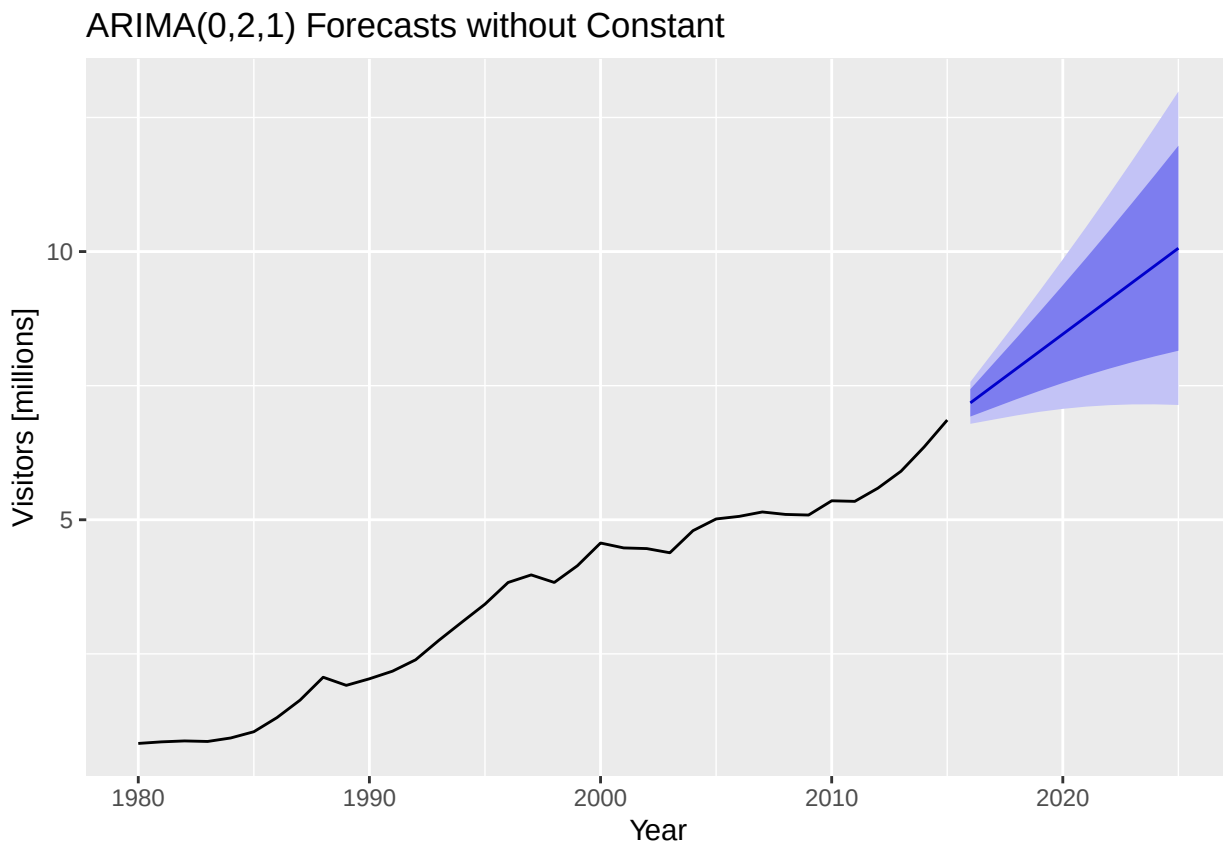
(e) Plot forecasts from an ARIMA(0,2,1) model with no constant.

ARIMA(0,2,1) with no constant

```
fit_austa_021 <- Arima(austa, order = c(0,2,1), include.constant = FALSE)
fit_austa_021
```

```
## Series: austa
## ARIMA(0,2,1)
##
## Coefficients:
##           ma1
##       -0.7263
## s.e.   0.2277
##
## sigma^2 = 0.03969:  log likelihood = 6.74
## AIC=-9.48   AICc=-9.09   BIC=-6.43
```

```r
# plot forecasts
fc_austa_021 <- forecast(fit_austa_021, h = 10)

autoplot(fc_austa_021) +
  labs(title = "ARIMA(0,2,1) Forecasts without Constant",
       x = "Year",
       y = "Visitors [millions]")
```



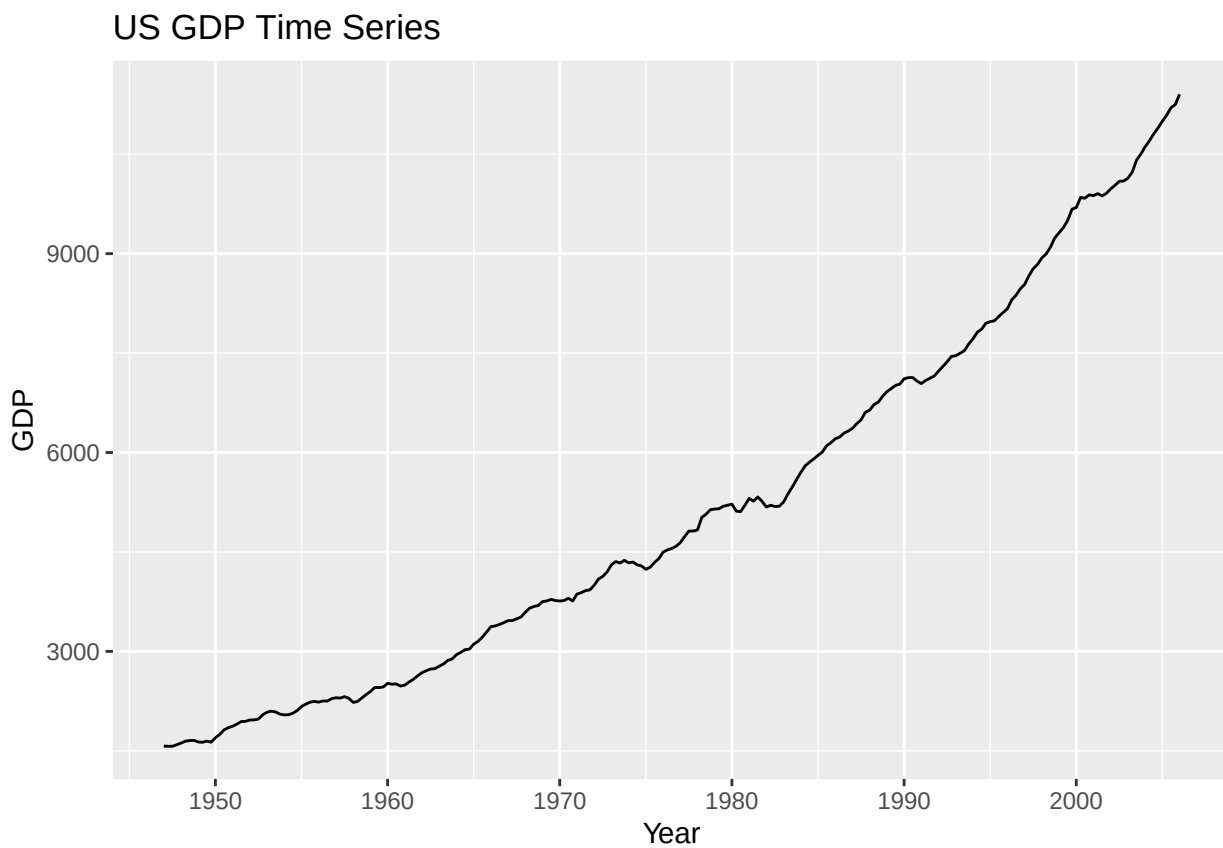
The ARIMA(0,2,1) model without a constant produces forecasts that continue the upward movement in the series. The prediction intervals widen rapidly, indicating increasing uncertainty in the future, as expected.

Question 9

For the usgdp series:

(a) if necessary, find a suitable Box-Cox transformation for the data;

```r
# plot to see behavior
autoplot(usgdp) +
  labs(title = "US GDP Time Series",
       x = "Year",
       y = "GDP")
```




```
# box cox parameter
lambda <- BoxCox.lambda(usgdp)
lambda
```

```
## [1] 0.366352
```

The plot of the usgdp series shows a strong upward trend and increasing variability over time, suggesting that a transformation may be useful to stabilize the variance.

The Box-Cox transformation parameter was estimated using BoxCox.lambda(), which gave lambda = 0.366. Since this value is between 0 and 1, a Box-Cox transformation is appropriate to reduce the increasing variance in the series.

(b) fit a suitable ARIMA model to the transformed data using auto.arima();

```
# using auto arima with previous calculated lambda
fit_usgdp <- auto.arima(usgdp, lambda = lambda)
fit_usgdp
```

```
## Series: usgdp
## ARIMA(2,1,0) with drift
## Box Cox transformation: lambda= 0.366352
##
## Coefficients:
##          ar1     ar2   drift
```

```
##       0.2795  0.1208  0.1829
## s.e.  0.0647  0.0648  0.0202
##
## sigma^2 = 0.03518:  log likelihood = 61.56
## AIC=-115.11   AICc=-114.94   BIC=-101.26
```

The auto.arima() function selected ARIMA(2,1,0) with drift and got the following coefficients: Estimated coefficients:

- AR(1) = 0.2795
- AR(2) = 0.1208
- Drift = 0.1829

This means that the GDP series required one difference (d = 1) to remove the trend, and the remaining dynamics are captured by two autoregressive terms. The drift term represents the long-term upward growth in GDP.

(c) try some other plausible models by experimenting with the orders chosen;

Trying ARIMA(1,1,0), ARIMA(0,1,1), and ARIMA(1,1,1)

```r
fit1 <- Arima(usgdp, order = c(1,1,0), lambda = lambda)
fit2 <- Arima(usgdp, order = c(0,1,1), lambda = lambda)
fit3 <- Arima(usgdp, order = c(1,1,1), lambda = lambda)

fit1
```

```
## Series: usgdp
## ARIMA(1,1,0)
## Box Cox transformation: lambda= 0.366352
##
## Coefficients:
##          ar1
##       0.6326
## s.e.  0.0504
##
## sigma^2 = 0.04384:  log likelihood = 34.39
## AIC=-64.78   AICc=-64.73   BIC=-57.85
```

```r
fit2
```

```
## Series: usgdp
## ARIMA(0,1,1)
## Box Cox transformation: lambda= 0.366352
##
## Coefficients:
##          ma1
##       0.4133
## s.e.  0.0453
##
## sigma^2 = 0.05519:  log likelihood = 7.4
## AIC=-10.79   AICc=-10.74   BIC=-3.87
```

```r
fit3
```

```
## Series: usgdp
## ARIMA(1,1,1)
## Box Cox transformation: lambda= 0.366352
##
## Coefficients:
##          ar1      ma1
##       0.9112  -0.5333
## s.e.  0.0461   0.1146
##
## sigma^2 = 0.03979:  log likelihood = 46.21
## AIC=-86.43   AICc=-86.32   BIC=-76.04
```

I tested ARIMA(1,1,0), ARIMA(0,1,1), and ARIMA(1,1,1).

The resulting AIC values were:

- ARIMA(1,1,0): AIC $= -64.78$
- ARIMA(0,1,1): AIC $= -10.79$
- ARIMA(1,1,1): AIC $= -86.43$

ARIMA(1,1,1) provided the best fit of the tested models, as it had the lowest AIC. However, the model selected earlier by auto.arima() (ARIMA(2,1,0) with drift - AIC $= -115.11$) still provides a better fit overall because it has a substantially lower AIC.

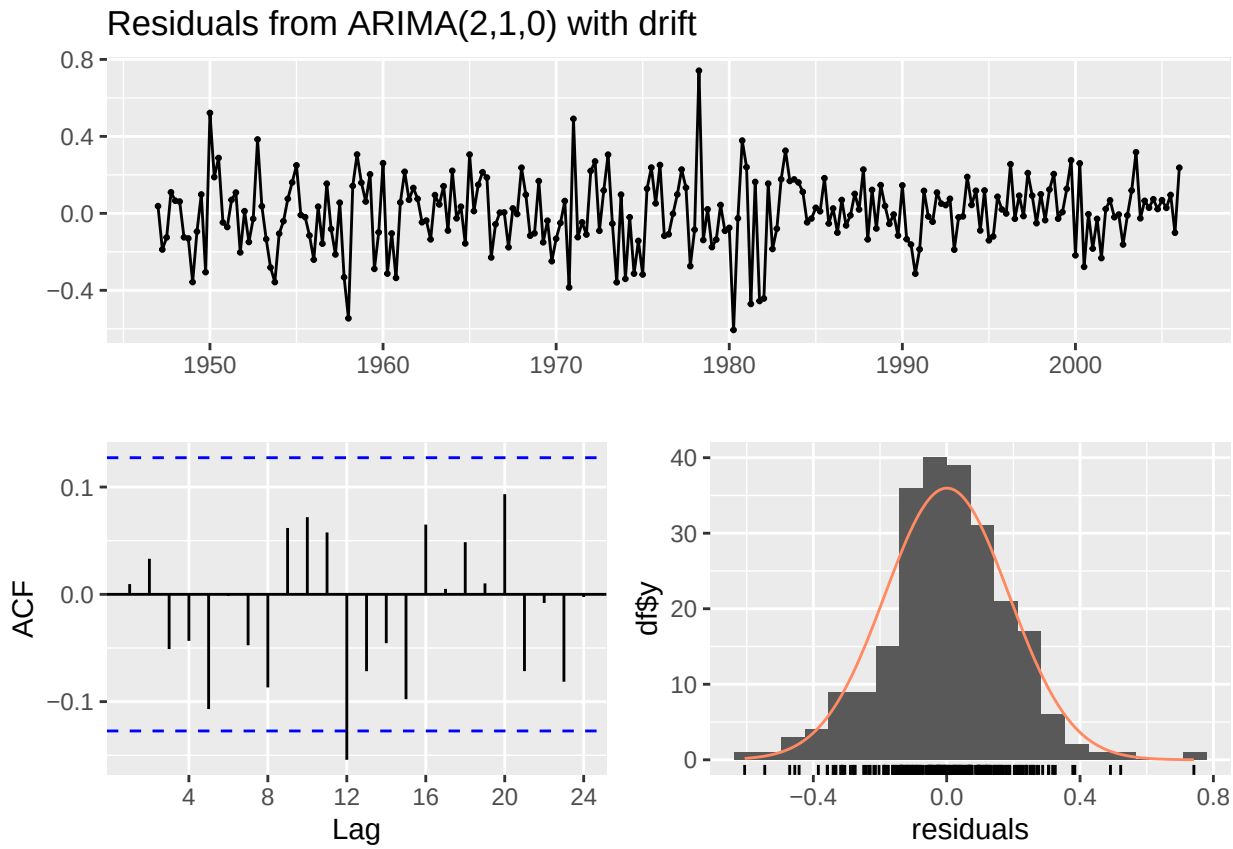
So, the ARIMA(2,1,0) with drift still is the preferred model for the transformed GDP series.

(d) choose what you think is the best model and check the residual diagnostics;

As explained above, the best model is ARIMA(2,1,0) since it has the lowest AIC.

Residuals for ARIMA(2,1,0)

```r
checkresiduals(fit_usgdp)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,1,0) with drift
## Q* = 6.5772, df = 6, p-value = 0.3617
##
## Model df: 2.   Total lags used: 8
```

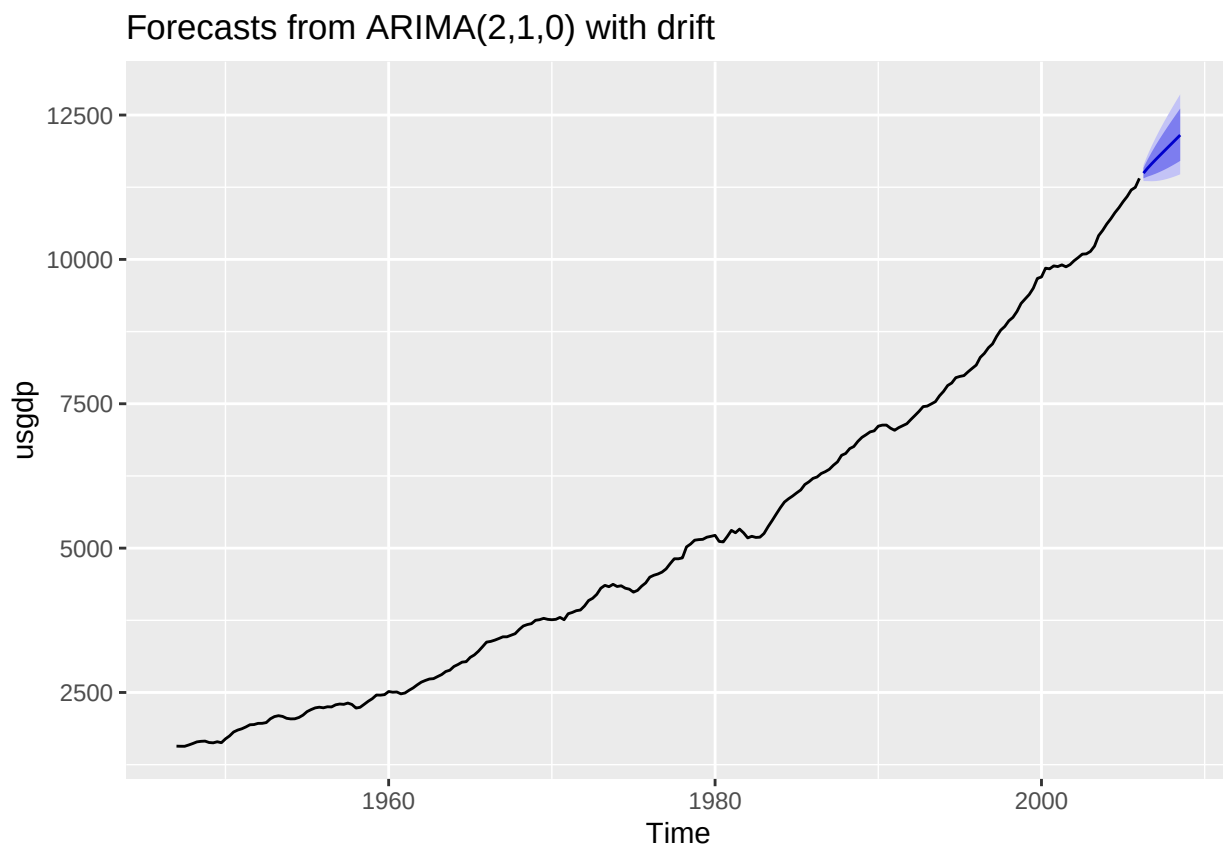
The residual time plot shows that the residuals fluctuate randomly around zero with no obvious patterns. The histogram of residuals appears approximately normally distributed. Also, the residual ACF shows no significant spikes outside the confidence bounds, which means that there is little remaining autocorrelation.

The Ljung–box test gives a p-value of 0.3617 (>0.05). Sp, we fail to reject the null hypothesis that the residuals are independent.

The residual diagnostics tells us that the ARIMA(2,1,0) with drift model provides an adequate fit to the transformed GDP series.

(e) produce forecasts of your fitted model. Do the forecasts look reasonable?

```r
usgdp_fc <- forecast(fit_usgdp, h = 10)
autoplot(usgdp_fc)
```



The forecasts show a continuation of the upward trend in US GDP. The prediction intervals are larger as the forecast horizon increases, showing a greater uncertainty in longer-term forecasts, as expected. The forecasts appear reasonable because they follow the long-term growth trend present in the historical GDP series and considering the greater uncertainty over time.

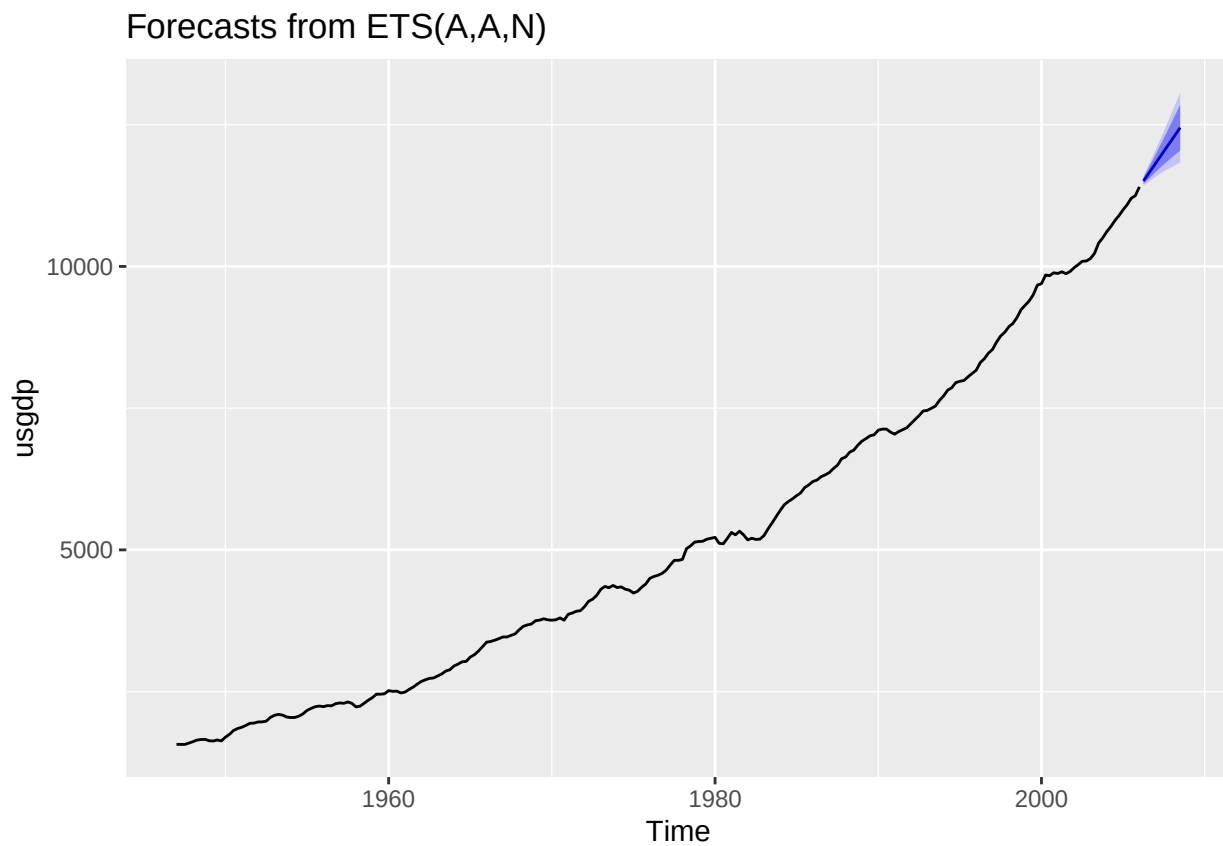
(f) compare the results with what you would obtain using ets() (with no transformation).

```
fit_ets <- ets(usgdp)
fit_ets
```

```
## ETS(A,A,N)
##
## Call:
## ets(y = usgdp)
##
##   Smoothing parameters:
##     alpha = 0.9999
##     beta  = 0.278
##
##   Initial states:
##     l = 1557.4589
##     b = 18.6862
##
##   sigma:  41.8895
##
```

```
##      AIC     AICc      BIC
## 3072.303 3072.562 3089.643
```

```r
usgdp_ets_fc <- forecast(fit_ets, h = 10)
autoplot(usgdp_ets_fc)
```



Using ets(usgdp), the selected model was ETS(A,A,N) (additive error, additive trend, no seasonality). The ETS forecasts also show a continued upward trend in US GDP, similar to the forecasts from the ARIMA(2,1,0) with drift model. Both models seem tocapture the strong growth pattern in the data, and the forecasts appear very similar. The main difference is that the ARIMA model used a Box-Cox transformation and differencing, while the ETS model directly models the level and trend components of the series.